Exploring the Correlation between Heat Exposure and Perinatal Depression

A Multi-Country Ecological Case-Crossover Study

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ABSTRACT

Background: While the impact of heat exposure on mental health has been well studied in the general population, little attention has been paid to perinatal women. Emerging research suggests an adverse effect, however, findings are only available from high-income settings.

Aim: This study aims to explore the correlation between heat exposure and perinatal depression (PND) in Bangladesh, Lesotho, Mozambique, and Nepal. The second objective is to study the lagged effect of heat exposure on PND in those countries.

Methodology: Data from the Demographic Health Survey (DHS) Program were bilinearly interpolated with NASA meteorological data. PND was defined as women with a Patient Health Questionnaire 9 (PHQ-9) score of ≥5, and heat exposure was defined as >50th percentile of daily maximum temperature. A time-stratified case-crossover study design with conditional logistic regression and distributed lag non-linear model (DLNM) was applied to answer research questions.

Results: 1,836 women screened positive for PND and were included in the analyses. A protective effect of heat exposure on PND was observed in Bangladesh. Daily maximum temperatures above the 50th percentile were associated with an increased likelihood of PND in Nepal and Mozambique, with delayed effects, suggesting a U-shaped relationship. In Lesotho, slightly elevated but non-significant values were observed at the 75th and 95th percentiles of maximum temperatures.

Conclusion: Heat exposure appears to be associated with PND, however, there is a need for further research, using longitudinal data, including additional environmental parameters and estimating mediating factors. Global health priorities should shift from the traditional focus on maternal mortality to the inclusion of mental health as a core indicator.

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ABBREVIATIONS

ANC ..... *Antenatal Care*

CI ..... *Confidence Interval*

DHS ..... *Demographic Health Survey*

DLNM ..... *Distributed Lag Non-Linear Model*

GC ..... *Geospatial Covariates*

GPS ..... *Global Positioning System*

HDI ..... *Human Development Index*

HIC ..... *High-Income Country*

IQR ..... *Inter-Quartile Range*

IR ..... *Individual Recode*

LMIC ..... *Low- and Middle-Income Country*

MDGs ..... *Millennium Development Goals*

MMH ..... *Maternal Mental Health*

NASA ..... *National Aeronautics and Space Administration*

OR ..... *Odds Ratio*

PCA ..... *Principal Component Analysis*

PHQ ..... *Patient Health Questionnaire*

PND ..... *Perinatal Depression*

SD ..... *Standard Deviation*

SDGs ..... *Sustainable Development Goals*

UN ..... *United Nations*

1 INTRODUCTION

Climate change represents one of the greatest public health challenges of our time, and rising global mean temperatures pose a significant threat to human health (WHO, 2023). The year 2024 goes down in history as the first year in which global average ambient temperatures surpassed the 1.5°C mark above the pre-industrial level. This indicates that the trend of rising average temperatures, including an increase in frequency and intensity of heat events, will continue as a result of anthropogenic climate change (Copernicus, 2025).

While the impacts of climate change on physical health – such as exacerbation of chronic illnesses and increased risk of transmission of infectious diseases (Liu et al., 2021; WHO, 2024) – are already well documented, little research has been conducted on the risks to mental health, particularly maternal mental health (MMH) (Barkin et al., 2025; Ulrich et al., 2025). This research gap is of particular concern given the significant global burden of MMH disorders and its implications for women and beyond.

Definition and Burden of Perinatal Depression

The global research gap on MMH stands in strong contrast to the high burden of mental disorders among perinatal women – defined as those who are pregnant or within one year postpartum (Howard et al., 2014). Global estimates suggest that approximately one in five women experiences a mental health condition during this time (Buchwald & Boudova, 2024). Alarmingly, suicide is one of the leading causes of maternal mortality, accounting for about 20% of postpartum deaths (Chin et al., 2022). Among perinatal women, perinatal depression (PND) is the most common morbidity reported (Barkin et al., 2024; Howard et al., 2014; Wisner et al., 2013). It is considered as part of major depressive disorder, with onset during pregnancy or within weeks after delivery, however, it is commonly extended to include the first year postpartum in both clinical and research settings (Bränn et al., 2024). PND is characterized by a range of affective, cognitive, and somatic symptoms, including persistent sadness, loss of interest in previously enjoyable activities, low self-esteem, sleep and appetite disturbances, anxiety, irritability, self-blame, and, in severe cases, hostility toward the infant and difficulty bonding. The burden is particularly high in low- and middle-income countries (LMICs). A meta-analysis including 23 LMICs estimated the prevalence at 25.5% (Roddy Mitchell et al., 2023).

Importance of Perinatal Mental Well-Being

Since MMH is a critical determinant of both maternal well-being and child development (Handa et al., 2024), the high burden of PND poses significant risks that extend beyond the individual, affecting families and societies (Carlson et al., 2025). Those affected by PND have been found to have an increased risk of recurrent mental illness, substance abuse and suicide attempts. This is reflected, for example, in

an up to five times higher risk of death due to suicide (Bränn et al., 2024; Chin et al., 2022; Roddy Mitchell et al., 2023). PND also has a negative impact on physical health and is associated with higher risk of pregnancy-related complications such as pre-eclampsia, gestational diabetes, longer hospital stays and a shorter breastfeeding period. Health consequences can also be of a long-term nature, for example through an increased risk of cardiovascular and autoimmune diseases (Bränn et al., 2024). The impact on neonatal and child health is equally severe, with children born to mothers with PND at higher risk for premature birth, low birth weight, low Apgar scores, cognitive and emotional delays, and long-term behavioral and developmental challenges (Roddy Mitchell et al., 2023; Wisner et al., 2013). Research also shows that PND is associated with insecure attachment and poor cognitive performance in the offspring. The implications for children can also be long-term, with increased susceptibility to mental health issues and cardiovascular diseases in adulthood (Roddy Mitchell et al., 2023). Additionally, MMH disorders impose a significant economic burden, contributing to higher healthcare costs, lost productivity, and long-term care needs, while also placing a strain on family relationships and social structures (Atif et al., 2015; Bränn et al., 2024). Given the far-reaching consequences, addressing PND through early intervention and support is vital for improving health outcomes for mothers and their children, as well as reducing the broader societal impact.

Health Policies and Health Care System

Therefore, ensuring adequate attention to PND is essential. Despite its high prevalence and severe acute and long-lasting effects, PND remains widely overlooked in global health agendas (Barkin et al., 2025). This neglect extends beyond healthcare systems to research and policy development, where MMH has historically been underprioritized (Handa et al., 2024). The Millennium Development Goals (MDGs), launched by the United Nations (UN) in 2000, prioritized reducing maternal mortality through Goal 5. While this agenda achieved substantial progress in lowering maternal mortality, it largely neglected the mental health of mothers (Engle, 2009). With the adoption of maternal and child health initiatives and the transition to the Sustainable Development Goals (SDGs), MMH continued to receive insufficient attention as maternal health programs have traditionally prioritized physical health outcomes, primarily focusing on maternal mortality as the key indicator (Handa et al., 2024). Despite growing recognition of MMH as a critical global health concern, it continues to receive little attention in research and policy (Ulrich et al., 2025).

Alongside the historical neglect of MMH, many health systems are not prepared to address the challenges posed by PND (KC & Kreyenbaum, 2025), particularly in LMICs. In these settings, healthcare systems often perform generally poorly, and perinatal health concerns remain significantly underemphasized, under-researched as well as under-resourced (Roddy Mitchell et al., 2023). MMH services have not yet been integrated into primary health care in many of these countries, because of multiple systemic barriers, including a lack of financial and human resources and the double burden of communicable and

non-communicable diseases. This consistently pushes mental health to the periphery of care priorities (Atif et al., 2015). Further, Handa and colleagues (2024) identify stigma as a major barrier to help-seeking, often discouraging women from pursuing treatment. This stigmatization further leads to a lack of routine mental health screening, resulting in a general lack of awareness and contributing to widespread underdiagnosis of PND. According to Carlson et al. (2025), up to 50% of PND cases remain undiagnosed. Additionally, mental health services are often separated from obstetric and pediatric care, resulting in missed early detection and treatment opportunities. Handa and colleagues (2024) further state that this also leads to breakdowns in provider communication and poor care coordination, leaving many women affected by PND without adequate treatment or support.

Risk Factors for Perinatal Depression

PND are influenced by a complex interplay of a range of biological, psychological, and social risk factors. During the perinatal period, women are particularly vulnerable due to rapid hormonal shifts, major life transitions, and increased caregiving responsibilities, all of which can place a significant burden on mental health (Barkin et al., 2024). In existing research, several well-established risk factors increase susceptibility to PND.

Biological risk factors include hormonal fluctuations, complications during pregnancy or childbirth, and underlying health conditions. Psychological and social stressors are equally influential, with well-established risk factors such as low perceived social support, financial stress as well as interpersonal violence contributing to PND (Roddy Mitchell et al., 2023). Exposure to domestic violence – physical, sexual, or emotional – has consistently been identified as a major determinant of MMH outcomes. Researchers found for example, that higher temperatures heighten interpersonal violence (Lawrance et al., 2022). This is closely linked to social support, as weakened social connections can impair the ability to process trauma and loss. A growing body of literature highlights the protective role that social support may have for perinatal mental health. For example, a recent study in Pakistan showed a strong association between low social support and postpartum depression (Sajjad et al., 2024). Further compounding this vulnerability are contextual stressors, such as poor access to healthcare, food insecurity, exposure to infectious diseases like HIV, or displacement due to conflict or disaster, all of which are common in many LMICs (Roddy Mitchell et al., 2023).

Heat Exposure as an Underexplored and Emerging Risk Factor for Perinatal Depression

Despite strong evidence of these risk factors for PND, one environmental determinant – heat exposure – remains underexplored (Barkin et al., 2024). With climate change increasing global ambient temperatures (Copernicus, 2025), the importance of environmental risk factors for mental health has started to gain attention (Liu et al., 2021), however, most research focuses on health impacts in child populations (Barkin et al., 2024). Perinatal women are often overlooked in this context, particularly their

mental health – despite the link to family and societal well-being and women's unique physiological and psychological vulnerabilities of perinatal periods (Barkin et al., 2024, 2025; KC & Kreyenbaum, 2025). To date, most of the research on heat exposure and maternal health investigates the physical health impact during pregnancy, including preterm birth, low birth weight, and stillbirth (Barkin et al., 2025; Dalugoda et al., 2022; Hanson et al., 2024; Kc et al., 2024; Wang et al., 2023). All of these outcomes have been associated with increased risks of postpartum depression, suggesting a potential pathway from heat exposure to PND (Barkin et al., 2025). Although there is a growing body of evidence on heat exposure and birth outcomes, the relationship with mental health in perinatal women is still poorly understood (Ulrich et al., 2025). Appendix 18 provides an overview of the suggested possible mechanisms underlying the correlation between heat exposure and PND, on the one hand through adverse birth outcomes and birthweight, on the other side through the experience of domestic violence and perceived social support. Additionally, housing conditions may act as a confounder, as it can determine the exposure to heat through insulation properties of floor, roof, and wall materials of houses, and has been associated with maternal depression (Corman et al., 2016). This directed acyclic graph (DAG) serves as theoretical framework to guide discussion about possible underlying mechanisms.

Emerging research, mostly from HICs, has started to study the link between heat exposure and MMH outcomes (Buchwald & Boudova, 2024). A study from North Carolina, United States, has shown that heat exposure is associated with increased rates of psychiatric emergency visits among pregnant women, especially during heatwaves (Ulrich et al., 2025). A longitudinal study from California, found that exposure to extreme heat during the last trimester of pregnancy was linked to higher odds of postpartum depression (Sun et al., 2024). Further, Runkle et al. (2024) explored the association between ambient temperature and psychiatric emergency visits during pregnancy and found that short-term high ambient temperatures lead to an increased risk of emergency department visits for maternal psychiatric outcomes. Lastly, a cohort study from China found that extreme heat may intensify maternal emotional and life-event stress on the day of exposure and up to two days afterward (Lin et al., 2017). While these findings yield important insight into the correlation between heat exposure and MMH outcomes, evidence from LMICs is scarce (Buchwald & Boudova, 2024). However, to fully explore this correlation, LMICs must be included in studies considering the high disease burden and vulnerability to climate risks. Buchwald and Boudova (2024) emphasize, that research in low-resource settings is possible; the authors cite three ecological studies that used Demographic and Health Survey (DHS) data and linked it with environmental risk factors to explore the association with adverse birth outcomes (Bachwenkizi et al., 2022; McElroy et al., 2022; Wang et al., 2023). While these are important contributions to closing the research gap, no studies have examined the effect of heat exposure on PND in LMICs.

This thesis addresses this knowledge gap by linking DHS data from four selected LMICs with temperature data extracted from the National Aeronautics and Space Administration (NASA) in an ecological case-crossover study to explore the correlation as well as lagged effects between heat exposure

and PND. Firstly, the aim of the thesis and the research questions are presented, followed by a description of the methodology used. The results are then presented, and findings are discussed and summarized. Finally, the paper elaborates on potential global health implications.

2 AIM

With the progression of climate change and the associated rise in both global average temperatures and heat events, assessing the association between temperature and MMH is fundamental for determining global and public health priorities as well as population risk. Therefore, this thesis examined the link between heat exposure and PND in women from four selected LMICs.

The primary aim of this thesis was to explore the correlation between heat exposure and PND in Bangladesh, Lesotho, Mozambique, and Nepal.

A secondary purpose was to assess the lagged effect of heat exposure on PND in selected countries.

Research Questions

Specifically, the following research questions will be addressed:

1. What is the correlation between heat exposure and PND in Bangladesh, Lesotho, Mozambique, and Nepal?
2. What is the lagged effect of heat exposure on PND in selected countries?

3 METHODS

This chapter describes the study design, setting, process of data management and cleaning as well as statistical analyses used to answer research questions of interest.

Study Design and Setting

In this multi-country ecological case-crossover study, the correlation between heat exposure and PND was explored. Based on data availability, the following four countries were selected: Bangladesh, Lesotho, Mozambique, and Nepal.

Bangladesh is a South Asian country with a population of 171.2 million, making it one of the most densely populated countries in the world. The country's Human Development Index (HDI) is 0.670 (UNDP, 2024) and Bangladesh is a middle-income country. The country has a humid, wet and tropical monsoon climate with one wet and one dry season per year. The average annual temperature between 1991 and 2020 was 25.49°C, with an average increase of 0.16°C per decade. Future climate projections indicate that this trend will continue, in addition to significantly higher heat risk by 2050. From 1971 to 2020, the country experienced an increase in the number of days exceeding 35°C by about 14 days per decade (The World Bank Group, 2024). Despite growing climate risk, MMH services remain limited. A significant treatment gap of 91.9% is driven by insufficient provider training, poor referral systems and a lack of structured guidelines (Dutta et al., 2024). Still, Bangladesh is among the first in WHO's South-East Asia Region to include maternal health in its 2020–2030 National Mental Health Strategic Plan, with measures like depression screening in maternal care and MMH training for health workers (Directorate General of Health Services, 2022). However, heat exposure is not yet recognized as a contributing risk factor.

Lesotho is a small, landlocked, lower middle-income country in southern Africa. The population is estimated to be 2.3 million in 2024, with an HDI of 0.520 (UNDP, 2024). Approximately 28% of the population lives in urban areas. Due to Lesotho's mountainous location, the climate can be described as temperate with alpine characteristics and an average annual temperature of 12.7°C between 1991 and 2020. However, since the 1960s, this value has increased by 0.76°C and this trend is expected to continue. Additionally, projections indicate that the frequency of heat waves will also increase, and the increase in the number of days with temperatures above 25°C has been identified as one of the most serious threats (The World Bank Group, 2021). Regarding MMH services, Lesotho has limited resources and programs in place to address the burden, which is estimated at 43% for postnatal depression. Although there are a number of primary health care services offering maternal and child care, routine screening for depression is not integrated into the spectrum of services (Mokwena & Makhozonke, 2024), and heat has not yet been recognized as a risk factor.

Mozambique is a low-income country on the southeastern coast of Africa with a population of 33.6 million. Approximately 38% of the population lives in urban areas and the country has an HDI of 0.461 (UNDP, 2024), indicating significant developmental challenges and making it one of the poorest countries globally (UN-Habitat, 2023). Due to its geographical location, Mozambique has a tropical and humid climate with one dry and one rainy season. The average annual temperature from 1991 to 2020 was 24.42°C. Projections indicate that this value could increase by 4.19°C by 2080-2099. This warming is projected to be more pronounced in inland regions, leading to more frequent and intense heatwaves (World Bank Group, 2023). Despite these risks, MMH services are severely underdeveloped. Mozambique's mental healthcare system focuses on primary care, however, significant gaps remain due to considerable staff shortages, with only 0.06 psychiatrists and 0.78 psychiatric beds per 100,000 inhabitants available (Gouveia et al., 2023). Specific programs or initiatives focused on MMH or in combination with heat as a risk factor have not yet been implemented.

Nepal is a small, landlocked lower-middle-income country in South Asia. The population is estimated at 29.7 million, with 20% living in urban areas and an HDI of 0.601 (UNDP, 2024). Due to its topographical characteristics, Nepal has a very diverse climate, ranging from subtropical lowlands to alpine highlands. As a result, climatic conditions vary greatly depending on the season and altitude. The average temperature drops from 24°C in the south to minus zero in the highest mountains. The historical temperature change between the periods 1900 to 1917 and 2000 to 2017 showed a temperature increase of 1.0°C to 1.3°C. This warming trend is projected to continue, potentially increasing the current median probability of heatwaves from 3% to 27% by 2090 (World Bank Group & Asian Development Bank, 2021). This increasing health threat is, however, not integrated into Nepali MMH services. As of 2024, there were no established services for screening and treating common mental health conditions in maternal health, even in tertiary level hospitals. This is of particular concern given that the prevalence of mental illness among Nepali women is estimated to be between 16 and 20%, and suicide has emerged as a leading cause of death among reproductive women in the country (Dhungana, 2024).

Study Population

As the DHS Program conducts women's health surveys among women of reproductive age, all study participants were between the ages of 15 and 45. For this thesis, women were considered eligible if they were in the perinatal period at the time of the DHS interview. This period includes women who are either currently pregnant or up to twelve months postpartum.

Data Source: Demographic Health Survey Data

Data on study participants and outcomes was derived from DHS datasets, which are nationally representative, cross-sectional survey data with a two-stage cluster-stratified sampling design. The DHS Program, launched in 1984 and primarily funded by the United States Agency for International

Development (USAID), is a global initiative that collects nationally representative, high-quality data to inform health and population policies in over 90 countries. The Program's main purpose is the advancement of collection, analysis and dissemination of population, health, and nutrition data. Next to these core areas, the DHS Program has traditionally focused on areas such as fertility, family planning, maternal and child health, and HIV/AIDS (Croft et al., 2023). Recently, the DHS has begun to expand to include emerging public health issues, including mental health. While these data remain limited in most DHS countries and waves, the most recent assessment (DHS-8 from 2018 to 2025) included a new optional module on mental health in the women's and men's questionnaires, covering topics such as symptoms of mental illness, care-seeking behavior, and previous diagnosis and treatment by a physician. One core part of this module screens for symptoms of depression using the Patient Health Questionnaire 9 (PHQ-9) tool, which is a validated psychometric scale (The DHS Program, 2022). Specifically, the nine items were integrated into DHS surveys to calculate a score based on symptom presence. Notably, even though the mental health module was integrated into DHS-8 in some countries, not all respondents of the sample answered these questions.

For the study population and outcome assessment, the Women's Questionnaires – coded as Individual Recode (IR) files – from the most recent DHS-8 were used, covering maternal health, mental health, and sociodemographic information, among others. Based on the availability of this module in combination with georeferenced coordinate data of cluster locations in the Geospatial Covariates (GC) files, a formal request for data access for Bangladesh, Lesotho, Mozambique, and Nepal was obtained through a study proposal. Estimated cluster centers were reported as longitude and latitude coordinates derived from a Global Positioning System (GPS), Census EA (Enumeration Area) boundary files, or public online maps or gazettes. To maintain participant confidentiality, the geographic coordinates of the clusters are randomly shifted by 2 to 10 kilometers, depending on whether they are urban or rural. These GC files can be linked to IR files through unique survey identifiers (Burgert, 2014).

For each country, the DHS-8 was conducted at different times; for Bangladesh, fieldwork was done from August to December 2022, for Lesotho from November 2023 to March 2024, for Mozambique from June 2022 to March 2023, and for Nepal from January to June 2022.

Data Source: National Aeronautics and Space Administration (NASA) Data

This paper uses daily maximum ambient temperature to assess heat exposure. This data was obtained from NASA's public Langley Research Center (LaRC) Prediction Of Worldwide Energy Resources (POWER) project, funded by the NASA Earth Science/Applied Science Program and established in 2003. The data was obtained from the POWER Project's Hourly V.2.4.5 version on 2025/02/15 with a spatial resolution of ½° by ⅝° of latitude/longitude grid (NASA Langley Research Center, n.d.). This application provides solar radiation and meteorological data for both single point locations and

rectangular areas as small as 2° by 2° and as large as 10° by 10° in longitude and latitude (*POWER | DAV*, n.d.).

Data Extraction, Management, and Cleaning

After formal data access was granted by the DHS Program, a three-step process of data management was performed. First, the countries' IR files were appended into a single dataset. Since the dataset from Nepal used the local bikram sambat (BS) calendar, dates were first converted into Gregorian calendar using a validated date conversion tool provided by the Nepali government. This step was necessary to ensure consistency in calendar format for the second merging process with daily temperature data from NASA.

Before, data cleaning included removing duplicates from the dataset and excluding participants who did not meet the inclusion criteria. During this process, first women who were not in the perinatal period were dropped. Definition of perinatal period was determined based on pregnancy and postpartum status. The latter was determined using the number of months since last childbirth, calculated by subtracting the child's date of birth (b3-01) from the interview date (v008). If this value fell between 0 and 12 months, the variable "postpartum" was coded as 1, indicating women were in the postpartum period. An eligibility variable ("eligible women") was created for this criterion, classifying women as eligible if they were either currently pregnant (v213 == 1) or within 12 months postpartum (postpartum == 1), to ensure the sample size consisted only of perinatal women.

For the second merging step, a rectangular sketching tool provided on NASA's POWER DAV website was used to fully cover areas of the four selected countries. For Nepal, Lesotho and Bangladesh, one rectangle was sufficient, whereas Mozambique required three rectangles. For the smaller countries, meteorological parameter files were merged into one comprehensive file per country. For Mozambique, the parameters were first merged within each rectangle, and then the datasets were appended across rectangles. For each country, the data downloaded ranged over the whole DHS fieldwork period, including 35 days before the first interview date. All merging and appending steps were performed in Stata 18. Daily data on five key meteorological parameters was extracted; Maximum temperature at 2 meters, minimum temperature at 2 meters, temperature range at 2 meters, wet bulb temperature (wbt) at 2 meters, and relative humidity at 2 meters.

Within each rectangle, NASA automatically provides requested parameters for multiple single-point locations. However, these points did not match the exact coordinates of DHS geocoded residential cluster locations. To spatially align the DHS GC dataset with NASA data points, Python's *cKDTree* package for bilinear interpolation was used to match each NASA data points to the nearest DHS cluster location provided in GC files. This ensured that the closest NASA locations were selected for each DHS cluster location. Any unmatched locations from NASA were automatically excluded, ensuring that the final

dataset contained only DHS clusters with the meteorological data from the nearest NASA location. This step was performed in the software Sypder 6.

Before the third and final merging step, a time-stratified case-crossover design was created. The choice for this study design is justified by its widespread adaptation in environmental epidemiology to assess short-term associations between environmental exposures and acute health outcomes (Tobias et al., 2024). Both heat exposure and PND fulfill these criteria. The design was introduced by Maclure in 1991 and is characterized by a within-subject comparison, where each subject serves as their own control (Carracedo-Martínez et al., 2010). The design automatically adjusts for time-invariant confounding variables, such as baseline mental health history, genetics, and socioeconomic factors (Carracedo-Martínez et al., 2010). Central to the design is its comparison of exposure levels during the case period (defined as shortly before the health event occurs) with exposure levels during at least one referent control period. If the control periods occur only prior to the health event, it is a unidirectional approach, whereas it is a bidirectional approach if control periods are defined both before and after the case period (Tobias et al., 2024). Notably, this study design is also reported to be well-suited for ecological-type exposures (Carracedo-Martínez et al., 2010). Since this thesis uses DHS data, which is cluster-stratified, and is combined with environmental data, the case-crossover design is appropriate (Carracedo-Martínez et al., 2010; Tobias et al., 2024).

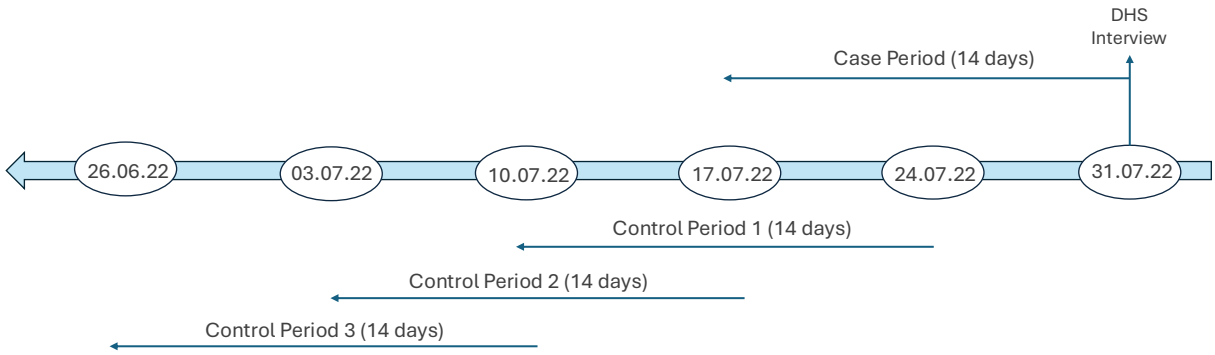

Figure 1. Example of Case-Crossover Design, showing the Case and Control Periods for One Participant Interviewed on July 31st 2022.

In this thesis, a unidirectional case-crossover study was established. This is because a 15-day case period (14 additional days prior to the date of the DHS interview) and three 15-day control periods, each separated by one week and occurring prior to the case period, were created. This was done through an expanding command in Stata 18, that assigned each day the corresponding date. The reference window was chosen based on items from the PHQ-9 that ask respondents, "*In the last two weeks, have you been feeling like ...*", resulting in a total of 56 days of environmental exposure data for each study participant. However, due to overlaps between the case and control periods, each study participant has a total of 35 unique days (see Figure 1). In this study design, participants screened positive for PND at the DHS

interview date, and negative for PND at the control periods. After creating the study design, the dataset was in long format and subsequently reshaped into a wide format for further analysis of lagged effects.

Finally, after the study design was created, the last merging step was performed based on unique cluster identifiers, country code and daily date that are consistent across GC and IR files, allowing the combined NASA datasets to be merged into the IR dataset. This resulted in a final dataset that includes both exposure and outcome data for all four countries. Since non-eligible women were already excluded when IR file was appended, DHS clusters that were not represented by any respondents, were not merged.

As part of data cleaning, missing data was handled by excluding women with incomplete data on the nine questions of the PHQ-9 scale. Responses coded as "7" (Refused) or "8" (Don't know) were treated as missing, and scores were only computed for participants with complete responses across all nine items. Moreover, variables that were not relevant to this thesis were deleted from the dataset.

Variables

The primary outcome PND is assessed by using the PHQ-9, which was used to screen survey participants for depression and provide an indication of symptom severity. The tool is part of the larger PHQ, that is used to screen for depressive and other common mental disorders. The PHQ-9 is a 9-item module for depression derived from the full PHQ, assessing depressive symptoms over the past two weeks and with points given (0 for "not at all", 3 for "nearly every day") to calculate a score that ranges from 0 to 27 (no depressive symptoms to major depressive symptoms). The score can also be classified into five categories to assess the severity of depressive symptoms: none-minimal (0-4), mild (5-9), moderate (10-14), moderately severe (15-19), and severe (20-27). Notably, since the ninth question assesses thoughts of self-harm and suicide, any positive response, regardless of frequency during the previous two weeks, requires further clinical evaluation. Therefore, a final diagnosis can only be made after a clinical examination and confirmation, emphasizing that the PHQ-9 is a screening tool and not a diagnostic tool (Kroenke et al., 2001). In a validation study, Kroenke and colleagues (2001) conclude that it is a valid and reliable tool for making criteria-based diagnoses and measurement of symptom severity for depression. For this thesis, PHQ-9 scores were recalculated using responses to mental health items mth8 through mth16. To support further analysis, symptom severity classifications were dichotomized into a binary variable indicating the presence of depressive symptoms, with "0" representing none-minimal and "1" including all higher severity levels. The assessment of relevant covariates included a range of socio-demographic, pregnancy-related, and interpersonal covariates that were binarily categorized. The following list shows relevant co-variates:

- Heat exposure: This is assessed as exposure to high ambient temperature, which is derived from daily maximum temperature from NASA. Temperatures above the 50th percentile are considered as heat exposure.

- PND: The outcome is expressed using the binary variable “depression”, which indicates participants are positive for PND if they have a PHQ-9 score of ≥5. This is derived from DHS.
- Age: Women were categorized into two age groups: 15–34 years and 35–49 years, based on their age at the time of the survey. The standard DHS variable v013 was used.
- Wealth: Household wealth was grouped into two categories using the combined DHS wealth index (v190): "Poor" (poorest and poorer quintiles) and "Rich" (middle, richer, and richest quintiles).
- Residency: Place of residence was assessed using the standard DHS variable (v025). Groups were “rural” and “urban”.
- Education: Educational attainment was grouped as "No or Basic Education" (no education or basic level) and "Secondary or Higher Education", based on standard DHS variable (v106).
- Housing Conditions: A binary index ("insulated" or "uninsulated") of household insulation quality was created based on their capacity to protect from heat.
- Antenatal Care (ANC): Adequacy of antenatal care visits was based on whether the respondent reported at least one ANC visit during pregnancy, following WHO’s minimum recommendation. The DHS variable (m14_1) on number of antenatal visits during pregnancy was used.
- Pregnancy Outcome: The outcome of the last pregnancy was categorized as "Live Birth" or "Non-live birth" (including stillbirths, miscarriages, and abortions), using DHS variable m80_1.
- Birthweight: Birthweight was grouped as "Low Birth Weight" (1500–2499 grams) and "Normal Birth Weight" (2500–3999 grams), using DHS variable m19_1.
- Domestic Violence: Experience of emotional, physical (less severe and severe), and sexual violence was assessed using standard DHS variables (d104, d106, d107, d108, respectively). These were considered individually to assess their distinct contributions to mental health outcomes. For analysis, the specific weight (d005) has been used since not all women from one household answered questions on domestic violence.
- Social Support: A proxy for social support was developed and separated into two categories (No/Low Support vs. High Support) based on median values.

Statistical Analysis

To review exposure of daily maximum temperature in each country, temperature patterns were plotted in R (version 4.4.3) using RStudio (2024.12.1). First, for each country, a daily average was computed by summarizing the daily maximums of each cluster. Missing values were excluded. These country-specific

daily averages of maximum temperature were then plotted into one figure to visualize the development across the whole study period for each country. Measures included mean, standard deviation (SD), minimum, maximum, median, and inter-quartile range (IQR), depending on data distribution.

During analysis, the individual sampling weight (v005) was applied to adjust for unequal probabilities of selection and to ensure national representativeness of countries. This weight accounts for the survey design and non-response at the individual level, making it appropriate for analyses of frequency proportions in descriptive statistics. As recommended by the DHS Program, this weight variable was divided by 1,000,000 prior to use in order to scale it appropriately. For retrieving p-values, significance levels the survey weight was used, as it accounts for the complex sample design by integrating the primary sampling unit (or cluster variable), stratification and weight variable (Croft et al., 2023). For assessing domestic violence, the weight (d005) was used. This step was done in Stata 18.

Principal Component Analysis (PCA) was performed to create variables for housing condition and social support systems of study participants. A binary index of household insulation quality was created using variables on floor, wall, and roof materials (v127, v128, v129, respectively). These materials were recoded as "insulated" or "uninsulated" based on their capacity to protect from heat. The first principal component was used to generate an insulation index, which was then dichotomized into "Insulated" and "Uninsulated" categories. Moreover, a proxy for social support was developed separately for each country using PCA on household and relationship variables (e.g., household head, co-residence with family members, decision-making power, and marital communication (v150, v136, v138, v504, v505, v743a)). The first component was extracted and divided into two categories (No/Low Support vs. High Support) based on median values. These were then combined into a single binary variable for the pooled dataset

Descriptive statistics, including frequency and percentage, were performed to summarize the characteristics of perinatal women by country and PND status. Then, significance testing using Chi-square test was performed to assess if differences in prevalence estimates of PND between variable groups were significant. Significance was determined using p-values ($p < 0.005$) and 95% Confidence Intervals (CI). This was performed in Stata 18.

The weighted prevalence of PND was calculated for each country as the proportion of individuals who screened positive for PND out of the sample size. These country-specific estimates were then summarized using meta-analyses to generate a pooled prevalence and to visualize cross-country variation. This was done in R (version 4.4.3) using RStudio (2024.12.1).

Exploratory analyses of the correlation between heat exposure and the likelihood of PND included conditional logistic regression models. Analyses were conducted first using a dataset in long format to see the association of a unit increase in exposure on the outcome, and then in wide format to investigate

the time-varying effect across lags. This was done separately for each country and pooled across countries. The outcome variable was PND ("depression"), and the exposure used was the daily maximum temperature of each case and control period. The models estimated odds ratios (ORs) with 95% CIs to quantify the strength and direction of the correlation between temperature exposure and PND. It is important to note, that conditional logistic regression models only takes participants who screened positive for PND into the model, thereby eliminating all study participants that did not screen positive for PND at the date of the DHS interview as they do not contribute to the likelihood (Tobias et al., 2024). In this thesis, the outcome variable of the conditional logistic regression is not the presence of PND per se, but whether a particular time period for the same individual is categorized as a case period (if PND occurred) or a control period (if it did not occur). In this way, the model can assess how changes in temperature exposure are associated with the occurrence of PND in the same individuals over time. These analyses were performed in Stata 18 and R (version 4.4.3) using RStudio (2024.12.1).

Since conditional logistic regression models assume a linear and immediate relationship between predictor and outcome, they may not adequately capture the complex dynamics of environmental exposures. To address this, the model was applied in a distributed lag non-linear model (DLNM) framework. DLNMs are well-suited for examining the environmental exposure and health outcomes (Gasparrini et al., 2010) since the effects are often delayed (Guo et al., 2013; Michelozzi et al., 2009) and non-linear, following a U-, V-, or J-shaped curve (Baccini et al., 2008; Curriero et al., 2002; Lyu et al., 2015; Ye et al., 2012). Unlike traditional epidemiological models that examine only exposure-response relationships, DLNMs integrate a second dimension – lag – allowing for simultaneous modelling of exposure and lag effects in time-series data. The definition of a *crossbasis* function, which combines two sets of basis functions – one for exposure-response relationship and one for the lag-response relationship – into a bi-dimensional space that determines the overall shape of the association (Gasparrini, 2011).

In this thesis, conditional logistic regression in a DLNM framework compares the exposure history – meaning the temperature across lags – within the same woman across different case and control periods and estimates how the risk of PND changes as a function of both temperature and time since exposure. The exposure-response relationship models how daily maximum temperature is associated with the risk of PND. A threshold function was used, assuming only temperatures above a specific cut-off influence the odds of PND, thereby imposing a one-sided effect. Temperature values were considered across a wide range, focusing on the 5th to 95th percentiles of daily maximum temperatures for each country, with adjustments made every 0.5°C. The median temperature for each country was set to provide a common reference point for estimating ORs. For the lag-response dimension, a natural cubic spline with three internal nodes was placed at equally spaced values over the 0 to 14-day lag period. This setting allows for flexibility while avoiding overfitting. The *crossbasis* function was then built to simultaneously

model exposure-response and lag-response dimensions to assess the full exposure-lag-response association between maximum temperature and PND.

From each country's DLNM, ORs and 95%CIs were extracted for (i) cumulative effects across the entire lag window, (ii) lag-specific effects, (iii) 3D-surface plots showing temperature by lag interactions, and (iv) slice plots showing ORs and 95%CIs by lag day. These were used to interpret both the intensity and temporal dynamics of temperature-related PND odds in the study sample. All DLNM analyses were performed using the *dlnm* package in R (version 4.4.3) using RStudio (2024.12.1).

Ethical Considerations

This study is based on publicly available, anonymized DHS data, and ethical approval for its use has been applied for. We consulted with the Swedish Ethical Review Authority regarding the use of the data; however, since the DHS data is anonymized, formal ethical approval was not required.

In terms of ethical considerations, privacy was maintained throughout the study. The data was securely stored and kept private and was not shared publicly.

4 RESULTS

This chapter describes the results of the analyses needed to answer research questions of interest. First, the study flow of participants, prevalence estimates of PND in a forest plot, and weighted descriptive statistics including chi-square tests are presented. Then, the average daily maximum temperature pattern of the four countries is described and then results of conditional logistic regression in Stata18 are presented, followed by each dimension of conditional logistic regression in DLNM framework.

Flow Chart of Study Participants

Of the 64,519 women interviewed during DHS-8 in Bangladesh, Lesotho, Mozambique, and Nepal, 10,021 remained in the dataset after dropping for perinatal status. Further cleaning for (complete) assessment of depression resulted in a sample size of 7,262 women. Bangladesh has 3,139 women (from 669 DHS clusters), Lesotho 390 (from 255 DHS clusters), Mozambique 2,896 (from 590 DHS clusters), and Nepal 838 (from 382 DHS clusters). Appendix 1 in the supplementary material visualizes the study flow of participants, and Appendix 2 shows spatial alignment of DHS clusters with NASA locations. Appendix 3 presents the number of NASA and DHS clusters for all included women and women that screened positive for PND.

Forest Plots for Prevalence Estimates

In a meta-analysis, the weighted prevalence of PND was assessed to ensure representativeness. Large variation in estimates across the four countries were observed (see Figure 1).

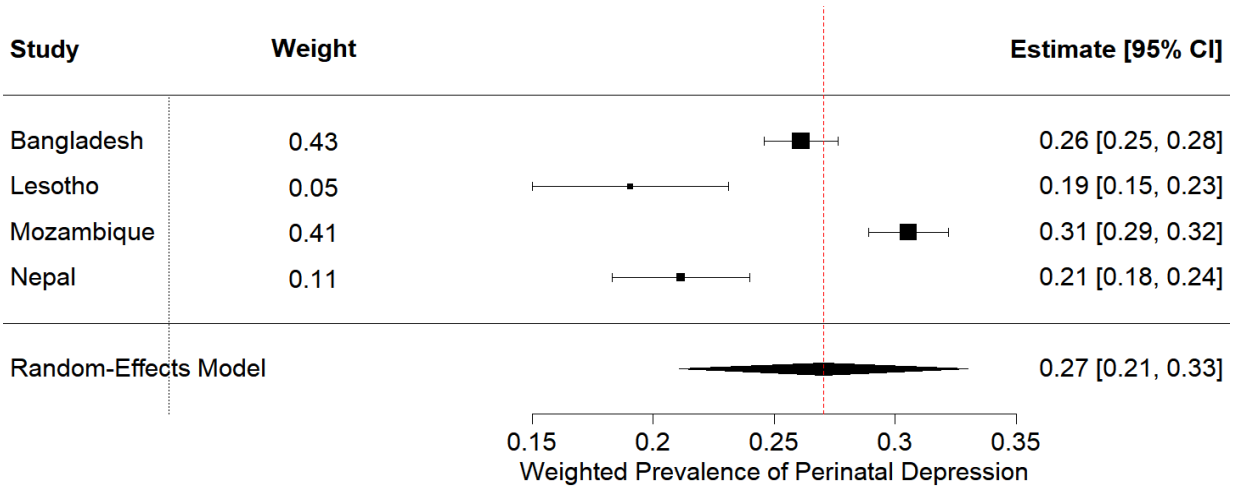

Figure 2: Forest Plot for Country-Specific and Pooled Estimate of PND.

The overall prevalence was 27% (95% CI: 21% to 33%), based on a random-effects model. Prevalence was highest in Mozambique (31%) and lowest in Lesotho (19%), with Bangladesh and Nepal reporting a prevalence of 26% and 21%, respectively. In the model, the weight varied significantly between studies

due to sample size differences. Larger sample sizes, such as those from Bangladesh and Mozambique, contributed the biggest weight to the pooled estimate (43% and 40%, respectively), while Nepal and Lesotho contributed only 11% and 5%, respectively.

Descriptive Statistics

Table 1a shows the weighted summary of socio-demographic characteristics of women with PND. A total of 7,339 women were included, with the majority from Bangladesh (43.4%) and Mozambique (40.8%), followed by Nepal (10.9%) and Lesotho (4.9%). In Bangladesh, Lesotho and Mozambique, the prevalence estimate for PND was higher in the 35-49 age group, while in Nepal, women aged 15-34 had higher estimates compared to the 35-49 age group. Chi-square test results showed that only in Mozambique was there evidence of a statistically significant difference in PND estimates between the two age groups. Women in lower wealth quantiles (poor) had higher PND estimates (ranging from 19.6% to 33.1%) in Lesotho, Mozambique, and Nepal compared to women in higher wealth quantiles. The p-value was statistically significant only in Mozambique (p-value: 0.037). Urban residence had slightly higher PND estimates than rural residence in all countries except Nepal, and no difference was statistically significant. Differences in women's educational attainment were not statistically significant in any country, but higher PND estimates were observed among women with no/lower education compared to women with basic/high education. Nepal is the only country where the PND estimate was higher for women with higher levels of education than for those with no/lower levels of education (22.5% vs. 19.5%). The index created to represent housing conditions showed that uninsulated conditions had higher PND estimates compared to insulated housing conditions in all countries except Lesotho. The difference was statistically significant in Mozambique (p-value: 0.001).

Table 1: Weighted Sociodemographic Summary Characteristics of Study Participants with PND..

	Bangladesh		**Lesotho**		**Mozambique**		**Nepal**	
Unweighted n (%)	3,139 (43.2)		390 (5.4)		2,865 (39.9)		838 (11.5)	
Weighted n (%)	3,182 (43.4)		362 (4.9)		2,996 (40.8)		799 (10.9)	
	Cases	p[b]	Cases	p[c]	Cases	p[b]	Cases	p[b]
Weighted n (%)	831 (26.1)		69 (19.2)		915 (30.6)		170 (21.2)	
Age group (years)[a]								
15-34	761 (25.5)		59 (18.9)		761 (29.4)		165 (21.5)	
35-49	69 (35.5)	0.061	11 (20.5)	0.605	155 (37.6)	0.014	4 (14.6)	0.357
Wealth [a]								
Poor	331 (23.7)		26 (19.6)		465 (33.1)		76 (22)	
Rich	500 (25.8)	0.654	44 (18.9)	0.786	451 (28.3)	0.037	93 (21.2)	0.686
Place of Residence [a]								
Urban	226 (26.9)		26 (19.3)		277 (32.5)		101 (19.3)	
Rural	604 (25.8)	0.673	43 (19.1)	0.920	639 (29.8)	0.283	69 (24.9)	0.108
Education [a]								
No/low	219 (28.3)		20 (23.9)		734 (31.7)		84 (19.8)	
Basic/ High	612 (25.4)	0.146	49 (17.7)	1.000	182 (26.7)	0.055	86 (22.8)	0.335
Housing Condition [a]								
Uninsulated	552 (26.5)		13 (14.9)		736 (32.5)		105 (22.5)	
Insulated	279 (25.3)	0.525	56 (20.6)	0.047	180 (24.5)	0.001	65 (19.5)	0.379

a: Number (%) presented.
b: Weighted p-value from Chi-square test, significance level = $p < 0.005$.
c: Unweighted p-value from Chi-square test, significance level = $p < 0.005$. Using weights resulted in no values for Lesotho.

Table 2: Weighted Pregnancy and Interpersonal Summary Characteristics of Study Participants with PND

	Bangladesh		**Lesotho**		**Mozambique**		**Nepal**	
Unweighted n (%)	3,139 (43.2)		390 (5.4)		2,865 (39.9)		838 (11.5)	
Weighted n (%)	3,182 (43.4)		362 (4.9)		2,996 (40.8)		799 (10.9)	
	Cases	p[b]	Cases	p[c]	Cases	p[b]	Cases	p[b]
Weighted n (%)	831 (26.1)		69 (19.2)		915 (30.6)		170 (21.2)	
ANC visits [a]								
Inadequate	44 (31.6)		5 (39.1)		90 (19.0)		6 (34.9)	
Adequate	482 (23.5)	0.057	46 (16.9)	0.386	612 (29.4)	0.920	109 (18.7)	0.141
Last pregnancy Outcome [a]								
Abortion, Stillbirth	41 (37.5)		6 (53.3)		14 (28.6)		7 (23.6)	
Live Birth	519 (23.9)	0.003	51 (18.0)	0.744	697 (29.3)	0.915	114 (19.0)	0.536
Birthweight [a]								
Low birthweight	52 (25.2)		4 (14.5)		42 (28.8)		10 (19.6)	
Normal birthweight	239 (21.1)	0.282	38 (16.9)	0.792	265 (23.4)	0.367	69 (17.6)	0.724
Perceived Social Support [a]								
High	n.a.		n.a.		429 (31.6)		84 (21.5)	
Low	n.a.	n.a.	n.a.	n.a.	347 (29.0)	0.314	85 (20.8)	0.825
Emotional Violence [a]								
No	n.a.		29 (14.4)		250 (28.6)		100 (19.9)	
Yes	n.a.	n.a.	24 (47.7)	0.000	994 (41.6)	0.007	24 (47.8)	0.001
Less Severe Physical Violence [a]								
No	n.a.		29 (15.2)		275 (30.5)		93 (20.1)	

Yes	n.a.	n.a.	25 (38.6)	0.002	70 (34.5)	0.397	41 (23.3)	0.007
Severe Physical Violence [a]								
No	n.a.		44 (18.8)		317 (30.6)		117 (21.7)	
Yes	n.a.	n.a.	10 (45.6)	0.091	28 (40.3)	0.142	17 (46.8)	0.007
Sexual Violence [a]								
No	n.a.		41 (17.5)		318 (30.5)		114 (20.9)	
Yes	n.a.	n.a.	13 (63.1)	0.001	27 (43.6)	0.124	21 (62.9)	0.001

a: Number (%) presented.
b: p-value from Chi-square test, significance level = $p < 0.005$.
c: Unweighted p-value from Chi-square test, significance level = $p < 0.005$. Using weights resulted in no values for Lesotho.

Table 2 shows the weighted summary characteristics of pregnancy and interpersonal characteristics for women with PND. Women who followed the WHO guideline on the recommended minimum number of ANC visits had lower estimates of PND in all countries except Mozambique, and no difference was statistically significant. A history of abortion, stillbirth or miscarriage for the last pregnancy was statistically significant for higher PND estimates (p-value: 0.003) among women in Bangladesh, but not in any of the other countries. Notably, Mozambique is the only country where the estimate of PND was higher among women who had a live birth. In Bangladesh, Mozambique, and Nepal, the estimates for PND were higher among women who had a low birthweight infant than among women who had a normal birthweight infant, but this was not statistically significant. The social support index (available only in Mozambique and Nepal) showed that PND estimates were higher among women with high social support than among those with low social support, but the differences were small and not significant. The DHS-8 assessed experiences of violence in Lesotho, Mozambique, and Nepal. The results show that women who experienced emotional violence had statistically significantly higher PND estimates (41.6% to 47.8%) than women who did not experience this form of violence. For less severe physical violence, women who experienced it had higher estimates than women who did not experience it in all three countries (23.3% to 38.6%), but this difference was statistically significant only in Lesotho and Nepal. Higher PND estimates (40.3% to 46.8%) were observed for women who experienced severe physical violence in all countries, with the difference being statistically significant in Nepal. Finally, the experience of sexual violence resulted in significantly higher estimates of PND (43.6% to 62.9%) compared to women who did not experience this form of violence. This was statistically significant in Lesotho and Nepal.

Assessment of Temperature Patterns Across Countries

The assessment (see Appendix 4) of the temperature patterns observed in the four countries, using average daily maximum temperatures across all clusters, shows considerable variation. Only clusters represented by PND-positive women are described, as only these women were used in exploratory analyses due to the nature of the study design. Appendix E shows the range, mean, median, IQR, and SD for maximum temperatures for each country.

Bangladesh shows a high mean maximum temperature (30.3°C) with low variability (SD = 1.9), indicating a relatively stable climate. Lesotho had a moderate mean maximum temperature (25.6°C) but a wide temperature range (14.2 – 37.2°C), and higher variability (SD = 4.7). In Mozambique, a mean maximum temperature of 31.1°C was observed, with substantial variation (SD = 4.0) and a broad range from 18.8 to 42.9°C. Nepal had the coolest mean maximum temperature (21.7°C) and the highest overall variability, as indicated by a wide interquartile range (IQR = 13.4). It also experienced the most extreme temperatures, with a minimum of −8.6°C and a maximum of 44.0°C. The values can be found in Appendix 5.

Results from Conditional Logistic Regression

Conditional logistic regression was performed to assess the linear relationship between daily heat exposure and PND, and only women with positive scores were included. Results from analyses using the long format dataset showed considerable variations in odds ratio and daily maximum temperature across countries. In Bangladesh, with every unit (1°C) increase in daily maximum temperature, the odds for PND statistically significantly decreased by 31.3% (OR = 0.69, 95% CI: 0.68 – 0.70, $p < 0.001$). In Lesotho, every 1°C increase in daily maximum temperature was associated with a 6.6% increase in the odds for PND (OR = 1.07, 95% CI: 1.04 – 1.09, $p < 0.001$). This was also statistically significant. In Mozambique, every 1°C increase in daily maximum temperature was associated with a 9.9% increase in the odds for PND (OR = 1.10, 95% CI: 1.09 – 1.11, $p < 0.001$). In Nepal, with every unit (1°C) increase in daily maximum temperature, the odds for PND statistically significantly increased by 41.4% (OR = 1.41, 95% CI: 1.38 – 1.45, $p < 0.001$). Across all four countries, every 1°C increase in daily maximum temperature statistically significantly heightened the odds for PND by 2.3% (OR = 1.02, 95% CI: 1.02 – 1.03, $p < 0.001$) (see Appendix 6, 7).

When the conditional logistic regression was performed in wide format of the dataset, the effect of daily maximum temperature at different lags on the outcome was assessed (see Figure 2, Appendix 8). ORs with 95% CIs were calculated for single day lags from 14 days preceding assessment day. The results for Nepal show that for every 1°C increase in daily maximum temperature at lag 1, the odds of PND increase by 1.29 (95% CI: 1.19 – 1.39). This increases until lag 11, where the odds peak at 1.56 (95% CI: 1.41 – 1.73), before decreasing to 1.45 (95% CI: 1.32 – 1.59) and then continuing to increase until lag 14. Thus, overall, in Nepal, the odds of PND increase significantly with each 1°C increase in maximum temperature at each lag, with the highest odds observed at lag 11. This suggests that the effect of maximum temperature on PND in Nepal is stronger if the heat exposure dates longer back (lag11 to lag 14). For Lesotho, results show that with every 1°C increase in daily maximum temperature at lag 1, the odds for PND increase by 1.07 (95% CI: 0.98 – 1.16). The odds remain relatively constant across the first lags (lag1 to lag7), ranging between 1.03 (95% CI: 0.95 – 1.12) and 1.08 (95%CI: 1.00 – 1.18). At lag 8 (1.11, 95% CI: 1.02 – 1.20) and lag 14 (1.11, 95% CI: 1.01 – 1.22), the odds peak and are statistically significant. At all other lags, the odds are above 1 but non-significant, suggesting a limited impact of maximum temperature on PND in Lesotho. In Mozambique, each 1°C increase in maximum temperature at lag 1 shows an increase in odds for PND by 1.14 (95% CI: 1.09 – 1.19). The odds remain consistently elevated the following lags, ranging from 1.09 (95% CI: 1.04 – 1.14) at lag 2 to 1.12 (94%CI: 1.07 – 1.17) at lag 7. Between lag 8 (OR = 1.11, 95% CI: 1.06 – 1.16) until lag 14 (OR = 1.07, 95% CI: 1.03 - 1.11) the odds fluctuate. Overall, consistently elevated and statistically significant odds for PND were observed with every 1°C increase in maximum temperature in Mozambique, suggesting an overall heightened risk over the reference period. In Bangladesh, with each 1°C increase in maximum temperature, a statistically significant reduction in odds for PND was observed across all lags. This

inverse association was strongest at lag 6 (OR = 0.61, 95% CI: 0.57 – 0.65), and odds were slightly increasing the following days. This suggests a persistent protective effect of the exposure to PND in Bangladesh.

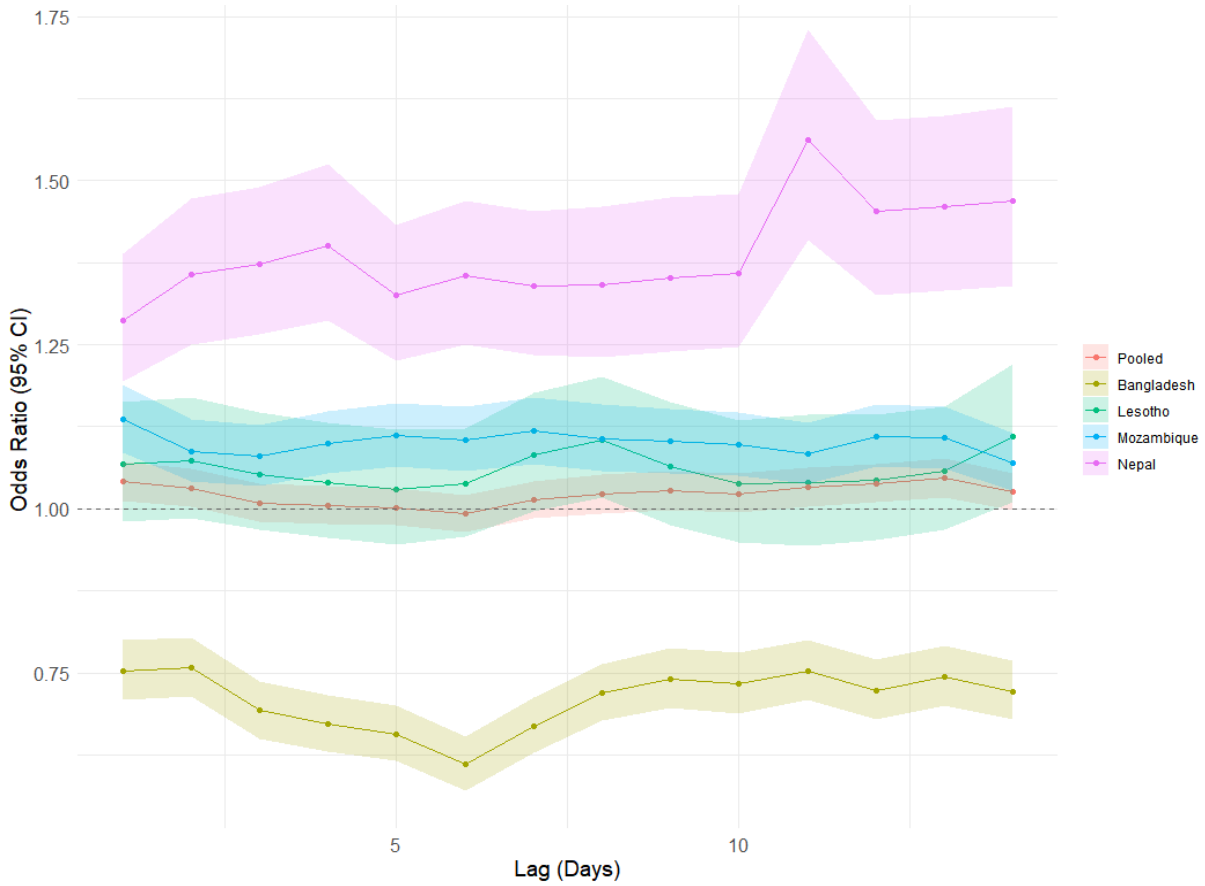

Figure 3: Results from Conditional Logistic Regression (Stata18) About Lagged Effect of Daily Maximum Temperature on PND by Country, including Pooled Estimate

In the pooled analysis of all countries, each 1°C increase in daily maximum temperature is associated with a modest but statistically significant increase in the odds of PND, particularly at short-term (lag 1-2) and longer-term lags (lags 9, 11-14), following a U-shape (not visible in Figure 2 though), supporting the notion of the delayed temperature effect on PND.

Results from Distributed Lag Non-Linear Model

Since conditional logistic regression models assume a linear relationship between the independent variable and the outcome, conditional logistic regression in a DLNM framework was performed to account for the delayed and non-linear relationship between exposure and outcome. This is necessary because the interpretation of the above results (except for Bangladesh) leads to the assumption that the lower the temperature, the lower the risk of PND. However, this is not the case, as shown by a body of research (Lin et al., 2017).

Assessment of Exposure-Response Relationship in DLNM

The first dimension of the DLNM establishes the exposure-response relationship between daily maximum temperature and PND. The cumulative effects across Bangladesh, Lesotho, Mozambique, and Lesotho can be seen in Figure 3. The y-axis shows the OR for PND, relative to the median maximum temperature (50th percentile). Daily maximum temperature is presented on the x-axis. The blue line presents the estimated cumulative OR at temperature points, and the shaded areas show the 95% CI. Specific point estimates at defined percentiles of maximum temperature can be found in Appendix 9-12.

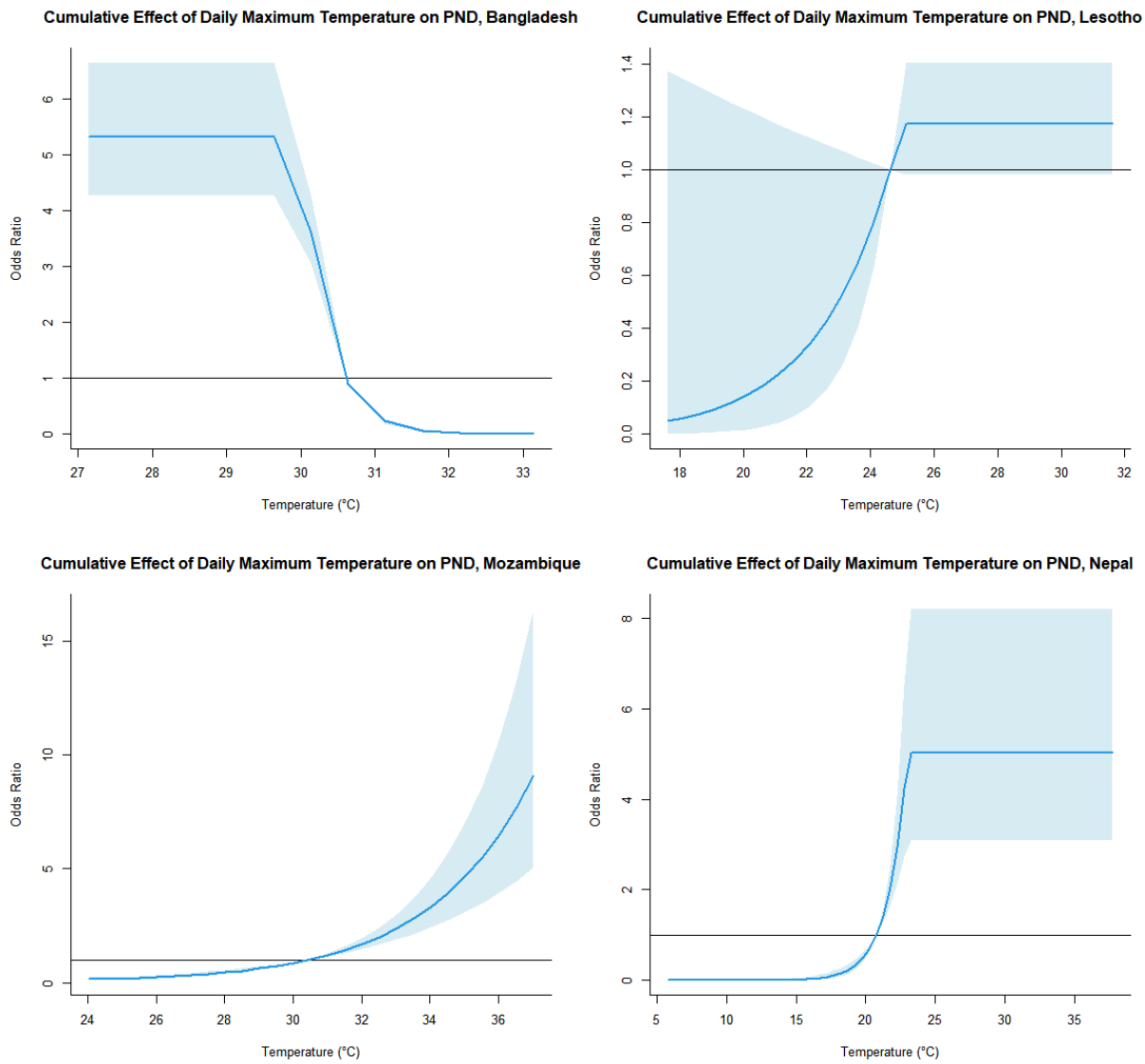

Figure 4: Temperature-PND Relationship Across Countries.

In Bangladesh, lower daily maximum temperatures show significantly higher odds for PND, with cumulative odds at 5.34 (95%CI: 4.28 – 6.66) at the 5th percentile (27.14°C), declining sharply around median temperatures (ref. = 30.6°C) to 0.0 (95%CI: 0.00 – 0.00) at the 95th percentile of maximum temperature (33.18°C). This suggests a potential protective effect of heat exposure. In Lesotho, an opposite pattern was observed. ORs of PND increased continuously with rising maximum temperatures,

from 0.05 (95%CI: 0.00 – 1.37) at the 5th percentile (17.62°C) to 1.19 (95%CI: 0.98 – 1.43) at the 75th percentile (27.57°C) and stagnating afterwards. While the odds at colder maximum temperatures showed a protective, but non-significant effect, maximum temperatures above the median (24.62°C) statistically significantly increased odds for PND. In Mozambique, the estimated cumulative effect of temperature on PND shows a similar pattern to Lesotho. Lower maximum temperatures at the 5th percentile (24°C) showed cumulative odds of 0.14 (95%CI: 0.09 – 0.24), indicating a protective effect. Once daily maximum temperature exceed the median of 20.44°C, the cumulative odds for PND was 2.51 (95%CI: 1.96 – 3.20) at the 75th percentile (33.54°C) and further increased with an accelerating slope to 8.68 (95%CI: 4.88 – 15.44) at the 95th percentile. In Nepal, a sharp increase in cumulative odds for PND was observed at temperatures above the median (20.7°C) reaching its peak at the 75th percentile (29.93°C) with a cumulative OR of 9.41 (95%CI: 4.88 – 18.14), not further accelerating. While this cumulative odd shows significant increase, uncertainty is rather high. Appendices 9-12 provide point estimates at the 5th, 25th, 50th, 75th, and 95th percentile of daily maximum temperatures for each country.

Assessment of Lag-Response-Relationship in DLNM

For assessing delayed effects, plots showing the lag-response dimension were created (see Figure 4). The x-axis shows the lag, meaning the number of days before the assessment date. On the y-axis, the OR including 95% CI is represented.

For Bangladesh, at 27.14°C, the ORs are highest at shorter lags (lag 1-2) and then reach their lowest values at lag 3-4, after which the ORs increase slowly but steadily before stagnating around lag 8 until the end of the reference window. Remarkably, the ORs and 95% CIs never fall below 1, indicating a statistically significant correlation between day and probability of PND at all lags. In Lesotho, at 25.12°C the ORs and 95% CIs follow a similar pattern, but are not statistically significant at any of the lags. The OR is highest at lag 0, then falls to its lowest value at lag 1, before reaching a second, but smaller peak at lags 3-5. From lag 8, the OR also stagnates until the end of the observation period. This indicates a non-significant correlation between day and probability of PND in Lesotho across all time points. The pattern observed in Mozambique is similar. At 25.02°C, the ORs reach their highest value at the interview date, fall to their lowest value at lag 1-2, before rising to their second highest value at lag 4-5. Thereafter, the ORs slowly decrease until the end of the observation period. It is noteworthy that only lag 0 has a positive OR, while all other lags have ORs below 1 and only lags 1-2 and 6-12 are statistically significant. This indicates that only these lags have a statistically significant correlation with the probability of PND, however, with protective characteristics. Finally, in Nepal, at 24.7°C, the ORs are highest for lags 0-2 and show a statistically significant increased odd for PND recent days. The lowest values are observed at lags 4-6, but are not significant. Thereafter, the ORs increase again up to lag 10 and then slowly decrease. Between lags 7-12, the ORs are statistically significant, which shows that in

addition to the acute lags, the lags between 7 and 12 are also statistically significantly correlated with the probability of PND in Nepal, indicating a delayed effect.

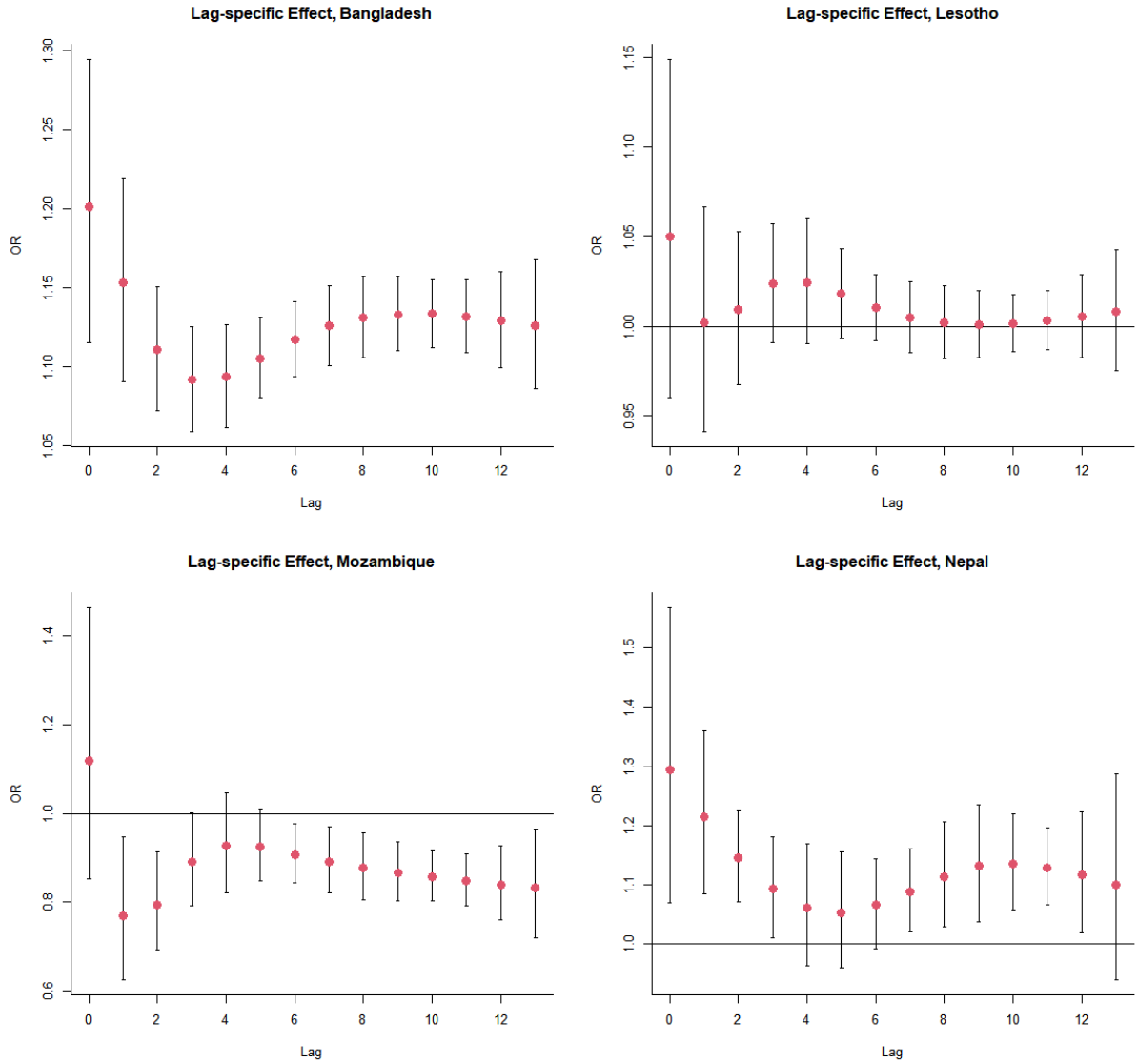

Figure 5: Lag-Specific Effect of Daily Maximum Temperature on PND, by Country

Assessment of Exposure-Lag-Response Relationship in DLNM

Figure 5 shows how different temperatures over time (lags/days) are correlated with the odds of PND in each country. The x-axis shows the observed daily maximum temperature, and the y-axis reflects the number of days between temperature and outcome. The z-axis shows the odds of PND relative to the centering temperature value. Point estimates and 95%CIs can be found in Appendices 9-12, and slice models visualizing the odds across for PND with increasing daily maximum temperatures at separate lags are shown in Appendices 13-16.

For Bangladesh, the exposure-lag-response surface indicates an inverse association between higher maximum temperatures and PND, suggesting a potential protective effect. The reference temperature is

set 30.6°C. Across all lags, the decreasing ORs are observed with rising maximum temperature. Lower temperature percentiles (e.g., 5th and 25th) are consistently associated with significantly elevated odds of PND. This effect intensifies over time, peaking around lags 7-13. In contrast, higher maximum temperatures above the 50th percentile show a protective effect, with ORs between 0.85 and 0.54 at the 75th and 95th percentiles (see Appendix 9). This plot suggests that colder maximum temperatures act as an acute and delayed stressor for PND, while heat exposure has a protective effect on PND.

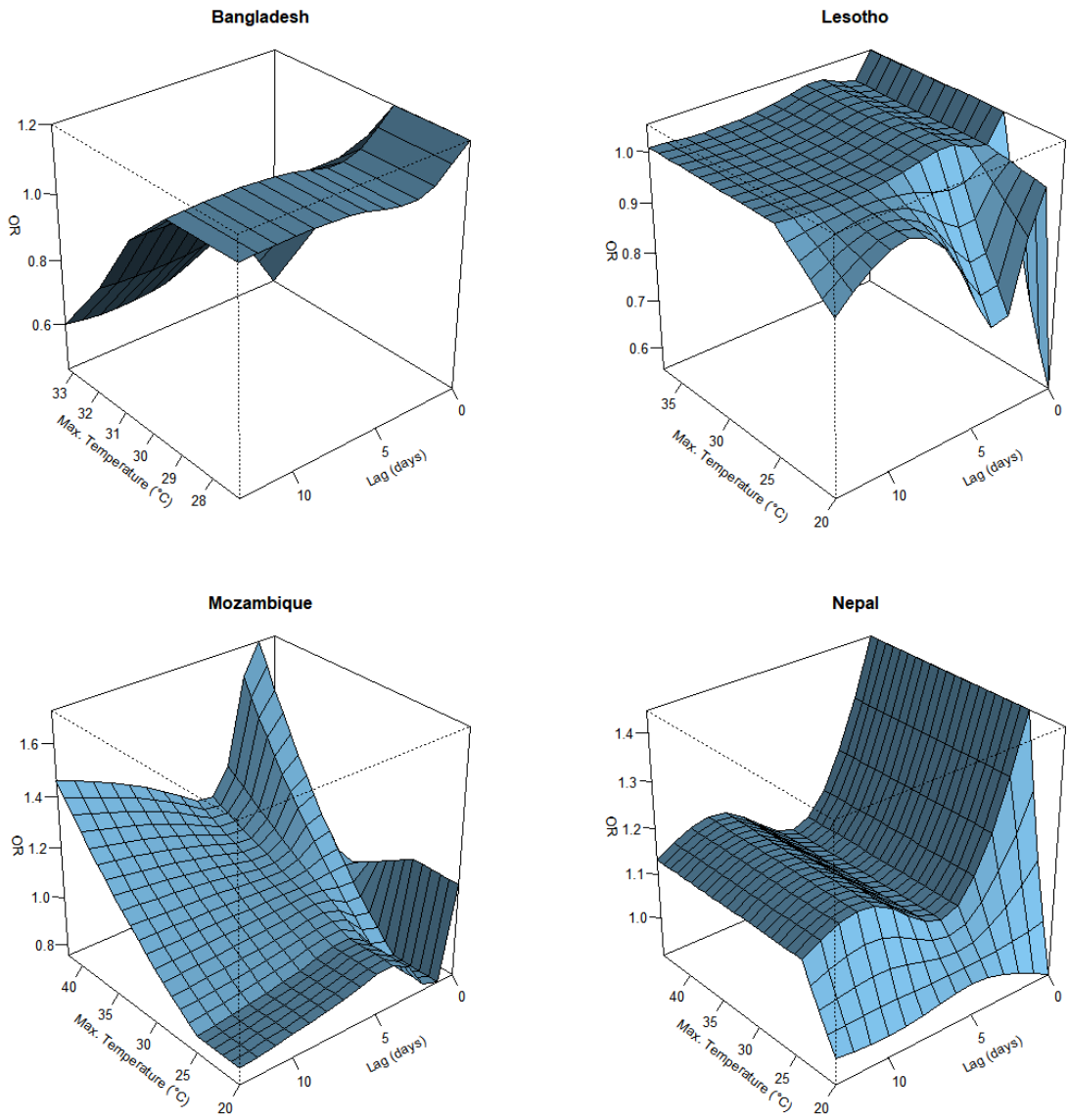

Figure 6. Results from DLNM for Correlation Between Daily Maximum Temperature and PND Accounting for Lagged Effects and Different Temperatures, by Country

In Lesotho, colder temperatures (e.g. 5th and 25th percentile) are associated with consistently lower OR across all lags, however with great uncertainty, compared with the median temperature at 24.62°C. Temperatures exceeding the reference (75th and 95th percentile), show slightly elevated odds for PND compared to the reference, particularly at lag 0 (see Appendix 10). With increasing lag, the effect

estimate stabilizes and fluctuates just above 1. Since CIs are not statistically significant, these findings suggest that the correlation between heat exposure and PND is weak in Lesotho.

In Mozambique, lower maximum temperatures between 14-27.5°C (5th and 25th percentile) show an association with reduced odds for PND, especially between lags 1-13. The reference temperature was 30.44°C. Higher maximum temperatures (75th and 95th percentile) are significantly associated with higher odds for PND after the day of assessment (lag 0). At the 75th percentile, effect estimates are most pronounced during shorter lags (lag 1-2), indicating an acute response. The 95th percentile shows the highest odds at lag 1 (1.37 (95%CI: 1.07 – 1.75). Following this peak, odds fall to 1.09 (remaining significant), before they rise again until lag 13 (OR = 1.24 (95%CI: 1.05–1.48)) (see Appendix 11). These findings suggest that there is a strong and significant correlation between heat exposure and PND, with an acute and delayed effect.

For Nepal, the reference temperature was 20.77°C. Colder maximum temperatures between 5.8 to 16°C (5th and 25th percentile) displayed a strong protective effect on PND, particularly at shorter lags (see Appendix 12). Heat exposure was observed to be statistically significantly linked to PND across most lags. For the 75th and 95th percentile, odds are identical and peak at lag 0 (OR = 1.45 (95%CI: 1.11–1.89)), then slightly declining to 1.08 (95%CI: 0.95–1.23) at lag 5, before they increase again up to 1.15 (95%CI: 1.02–1.31) at lag 12 (see Appendix 12), suggesting a delayed effect.

5 DISCUSSION

This chapter summarizes the main findings and relates them to previous work in this area, highlighting areas of agreement and disagreement with my findings. It also highlights the novelty of this thesis and discusses its strengths and limitations. Finally, the findings will be placed in a local country context.

Summary of Key Findings

This thesis examined the correlation and lagged effects between heat exposure and PND among 1,836 women who screened positive for PND in Bangladesh, Lesotho, Mozambique, and Nepal. The DLNM showed country-specific and time-dependent effects: In Bangladesh, a consistent protective effect of heat exposure was observed, with colder maximum temperatures increasing the risk of PND. In contrast, Nepal had the highest heat-related odds of PND, with both acute and delayed effects. Mozambique showed a strong, persistent, statistically significant association between heat exposure and PND. Lesotho showed small and non-significantly increased odds of PND with heat exposure, indicating no strong association.

Relation to Previous Work

Previous studies exploring the intersection of heat exposure and MMH have generally reported adverse effects of heat exposure, with various study designs employed and mental health outcomes investigated. However, most of this research has been conducted in populations from HICs and specifically focusing on heat-related emergency department visits in pregnant women.

For instance, Sun et al. (2024) conducted a retrospective cohort study in California and found that postpartum exposure to extreme heat was linked to increased odds of postpartum depression. However, their analyses were restricted to the postpartum period, not assessing potential heat exposure during pregnancy. Similarly, in their study, Runkle and colleagues (2024) focused on pregnant individuals and identified a strong association between acute exposure to hot temperatures and increased psychiatric emergency visits in North Carolina. Lin et al. (2017) investigated pregnant women in Shanghai and found a U-shaped relationship between temperature and maternal stress. Another North Carolina study examined mental health disparities among pregnant women during warm seasons from 2011 to 2019. Using emergency department data, the authors found a 13% increase in severe mental illnesses after acute heatwave exposure and a 37% rise with prolonged heatwaves. While the study reported no significant link between heatwaves and Perinatal Mood and Anxiety Disorders (PMAD) – defined as reoccurrence of existing depression or anxiety – a protective, but nonsignificant effect was found during acute heatwave exposure (Ulrich et al., 2025). These findings align with the mixed results observed in this thesis.

First, the findings from Lesotho, Mozambique, and Nepal indicate a correlation between heat exposure and PND and agree with results from Sun et al. (2024), Runkle et al. (2024), and Lin et al. (2017). In these countries, higher odds for PND at shorter lags and longer lags with heat exposure were observed (except Lesotho). This harvesting effect has also been reported in other studies. Findings from Bangladesh, on the other hand, align with results from Ulrich and colleagues (2025), indicating a protective effect of heat exposure on PND.

Novelty of Thesis

These mixed findings provide novel input for this area of research. Studies identified through a systematic literature search (Appendix 17) have mainly focused on HICs, while this thesis examined perinatal women four LMICs from South-East Africa and South-East Asia. Buchwald and Boudova (2024) highlight in their paper that in order to holistically assess the impact of heat exposure on MMH, the scope of research needs to be extended beyond high-income settings to include understudied LMICs. This is particularly important as the effects of climate change and expected temperature increases will be most pronounced in these settings. In combination with a lack of adequate infrastructure for mental health services (Sharpe & Davison, 2021), the vulnerability for PND is possibly much higher than in HICs. Further, while most studies look either at pregnant or postpartum women, this thesis included both groups in the analysis, providing a more nuanced perspective on perinatal populations. Lastly, some papers (Runkle et al., 2024; Ulrich et al., 2025) assessed outcomes based on emergency department visit data, which may capture rather severe or clinically emergent cases of mental issues, therefore not reflecting the broader burden of MMH issues in the population and limiting generalizability to other groups. In comparison, this thesis leveraged nationally representative DHS survey data to screen for PND, offering nation-wide insights.

In doing so, it was also methodologically novel; linking DHS mental health data with NASA temperature data was, to my knowledge, a new approach to studying this intersection. Previous research has linked DHS data with temperature data, but the outcomes examined included adverse birth outcomes (McElroy et al., 2022; Wang et al., 2023). In addition, methodological novelty was offered using conditional logistic regression within a DLNM framework. Unlike Sun et al. (2024) and Runkle et al. (2024), who did not apply DLNM methods, this thesis assessed both non-linear and delayed effects of heat exposure on PND. Two published papers, Lin et al. (2017) and Ulrich et al. (2025), applied DLNM, however, key analytical limitations remained. Ulrich et al. (2025) defined heatwave events as several consecutive days of elevated temperature above a certain threshold. Their analyses therefore relied on binary heatwave indicators and modeled lag effects only up to seven days, potentially overlooking more cumulative or delayed impacts of temperature. By contrast, this thesis modeled daily maximum temperature as a continuous exposure and examined lagged effects up to 14 days, allowing for a prolonged and more sensitive assessment. Lin et al. (2017) used daily average temperature and temperature range as exposure

metrics, which may cover the effects of extreme heat peaks and temperature variance more relevant to acute stress or depressive responses. However, the lags included were limited to eight, making detection of prolonged effects impossible. Additionally, their outcome measure – maternal emotional stress – was measured using the SCL-90-R, which includes a subscale for depression, but primarily captures general psychological distress (Lin et al., 2017) rather than diagnosing or screening for PND specifically.

Possible Underlying Mechanisms

Studies that identified a correlation between heat exposure and PND suggest a potential pathway through biological, psychological, and social mechanisms. Appendix 18 visualizes hypothesized pathways.

Biologically, heat exposure may affect mental health through physiological changes such as alterations in blood flow or serotonin levels (Thompson et al., 2023). Because of their physiological status, pregnant and postpartum women are particularly vulnerable to heat (KC & Kreyenbaum, 2025). As Ulrich and colleagues (2025) state, the underlying biological mechanisms by which heat exposure worsens perinatal mental health outcomes are still unclear, however, there is scientific consensus that a comprised ability to thermoregulate core body temperature increases women's vulnerability to high ambient temperatures. Possible pathways include neuroendocrine dysregulation through elevated cortisol levels, which contributes to poor stress response, and a dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis, resulting in altered cortisol rhythms (Sun et al., 2024). Further, heat stress may comprise blood-brain barrier (BBB) integrity, leading to the infiltration of immune cells, inflammatory cytokines and reactive oxygen species into the brain, which is associated with psychiatric symptom exacerbation. Additionally, exposure to high ambient temperatures may also interfere with certain psychiatric medications that impair heat dissipation. As a consequence, deteriorations of mental health diseases increase in likelihood (Ulrich et al., 2025).

A potential mediation pathway also exists via pregnancy complications and adverse birth outcomes, which are consistently associated with heat exposure and are established risk factors for PND (Barkin et al., 2025). A meta-analysis found that during heatwaves, the odds for any obstetric complication increased by 25%. Additionally, with every 1°C increase in heat exposure, the odds for preterm birth rose by 1.04 (95%CI: 1.03 – 1.06), for stillbirth by 1.13 (95%CI: 0.95, 1.34), for congenital abnormalities by 1.48 (95% CI: 1.16 – 1.88), and for gestational diabetes by 1.28 (95% CI: 1.05 – 1.74) (Lakhoo et al., 2025). This is in line with findings from Zhu et al. (2024), who reported a 40-41% increased risk of low birth weight in East Africa and South Asia with heat exposure during pregnancy, and Kc et al. (2024) found a 29% higher risk of preterm birth over 28 days in Nepal at higher ambient temperatures. Combining these findings with the results from a recent meta-analysis that found positive and statistically significant associations between birth complications and PND symptoms (Cárdenas et al., 2025), there is reasonable assumption that these factors may act as mediators linking heat exposures to PND.

In addition to biological pathways, psychological mechanisms have also been proposed in published literature as possible mechanisms. This includes, disrupted sleep patterns as a result of elevated ambient temperatures, which have been shown to impair cognitive functioning and emotional regulation that possibly exacerbates mental health vulnerabilities (Thompson et al., 2023; Ulrich et al., 2025).

Among the possible indirect pathways of heat exposure and PND are social and socioeconomic factors. Thompson et al. (2023) and Lawrence et al. (2022), for instance, report that societal changes may occur as a result of non-optimal ambient temperatures, resulting in heightened aggression and stress from reduced agricultural and economic outputs. One study demonstrated, for instance, that the association between heat exposure and postpartum depression is strongest following violent experiences (Gastaldon et al., 2022). This is closely related to social support systems that provide resilience in the face of distress. A study from Pakistan reported a significant association between social support and the development of postpartum depression, emphasizing the role for MMH (Sajjad et al., 2024). Disruption of healthcare and employment may be another consequence of heat exposure, potentially taking a significant toll on MMH (Barkin et al., 2024). Furthermore, climate change is not only driving up ambient temperatures but is also causing growing numbers of people to be displaced. It is estimated, that 80% of this group will be women, further compounding their vulnerability due to housing instability and lack of support systems (Barkin et al., 2024, 2025). These increased risk factors for PND, in combination with social disadvantages in the context of climate change, including limited access to basic knowledge and adaptation resources, and a lack of appropriate perinatal mental health services, put perinatal women at increased risk for PND (Roddy Mitchell et al., 2023; Zhu et al., 2024).

Strengths

This thesis has several strengths and limitations that are important for translating the research findings into broader contexts.

First, a key strength of this thesis is its novel linkage of DHS data with high-resolution gridded temperature data from NASA. This enabled both spatially and temporally accurate as well as objective measurement of heat exposure and supported a deeper understanding of its correlation with PND. Another strength is the focus of four LMICs. As previously mentioned, no study has yet explored the correlation between heat exposure and PND in these settings, leaving a fundamental research gap. This thesis targeted that gap by covering LMICs from South-East Asia and South-East Africa. Furthermore, the use of large-scale DHS data from nationwide surveys in combination with appropriate weights for representativeness ensured that research findings can be generalized to national populations. The methodology applied in this thesis represents another major strength: The combination of conditional logistic regression with DLNM allowed a nuanced analysis of exposure-lag-response relationship. Unlike other studies using DLNM at the intersection of heat exposure and MMH (Lin et al., 2017; Ulrich

et al., 2025), this thesis captured the impact over a 14-day lag period and of continuous temperature exposure rather than using categorical heatwave classification. This provided important insights into temperature-related gradients and lagged correlations of heat exposure and PND, offering input for temperature thresholds and critical time windows for each country. In addition, due to the study design and its characteristics of within-subject analyses, any potential risk of confounding was eliminated by controlling for the influence of risk factors for PND.

Limitations

The choice of this study design, however, also yields significant limitations for my thesis that need to be discussed. On the one hand, the case-crossover design, paired with conditional logistic regression, is well-suited for exploring short-term relationships and allows for simultaneous integration of time-varying effect modifiers (Tobias et al., 2024). On the other side, this design assumes the availability of time-varying outcome and covariate data across multiple time points for each individual. This assumption conflicts with the cross-sectional nature of DHS data, where all information on the outcome and covariates is collected at a single time point during the survey interview. As a result, temporal variation in variables within one individual between case and control periods cannot be captured. This limits the full potential of the case-crossover design, including the assessment of effect modification by time-varying covariates. Additionally, as the case-crossover design relies on within-person variance across strata, interaction terms with time-invariant variables cannot be estimated in conditional logistic regression models. Such variables are treated as constant, making their effect modification impossible and representing a core methodological limitation of using a case-crossover design with cross-sectional data. Consequently, as PND and the covariates are assessed only once, the ability to track changes and ensure proper temporality is limited.

Another limitation of the cross-sectional nature of DHS data is the assessment of PND. The PHQ-9 was used to screen for presence of depressive symptoms two weeks preceding the DHS interview date. If a women screened positive, in the study design, it is assumed that she was negative for PND during the control periods. However, it may be possible that those respondents have been suffering from depressive symptoms long before the DHS interview. So, the static measurement of the outcome introduces uncertainty about when the depressive episode occurred in relation to the exposure. Alongside this constraint comes a limited – but potential – risk of recall bias. This has been reported in a study using DHS data (Zhu et al., 2024), since the questionnaires collect data retrospectively. Since PND is considered an acute outcome, and only two weeks prior to the interview date are assessed, this risk is limited but present.

For the outcome, further limitations need to be considered. PND was classified based on PHQ-9 score solely. In the methodology chapter, this psychometric scale has been stated as valid, however, the tool

does not provide clinical verification. Notably, the ninth item of the questionnaire must be flagged separately for clinical interpretation, as it asks about respondents' thoughts on self-harm and suicidal ideation (Kroenke et al., 2001). Therefore, there is risk of misclassification of women based on the PHQ-9 score solely. Additionally, the scale is designed to screen for depressive symptoms in the general population, without a special focus on perinatal populations. Other scales, such as the Edinburgh Postnatal Depression Scale (EPDS) – although targeted at postpartum women – might provide more accurate estimates of PND.

Despite established evidence of the impact of various environmental factors on mental health, including air pollution, humidity, and biodiversity (Thompson et al., 2023), only heat exposure was included into this thesis. Assessing such factors in combination with heat exposure may result in more accurate effect estimates. Moreover, the assessment of heat exposure also displays notable limitations. For instance, DHS locations of individuals are randomly shifted between 2 to 10 kilometers to ensure their privacy and then bilinearly interpolated to match NASA cluster locations, thereby distorting the actual temperatures experienced by individuals. Additionally, the maximum temperature was assigned to respondents at the cluster level, implying that individuals within one cluster are uniformly exposed to the same temperatures at the same time. However, there is variation in respondents' capacity to protect from and adapt to heat exposure, that cannot be acknowledged. This lack of information further hides the true heat exposure experienced by participants. Related to the availability of temperature data on the cluster level, it is also important to consider the risk of ecological fallacy, implying that conclusions about correlation and lagged effects can only be made on the population level (of perinatal women) not on the individual level.

Furthermore, some covariates considered as relevant risk factors for PND were limited by data availability. This includes, for example, respondents' HIV/Aids status, which was not available in DHS IR files. However, this is considered as a strong predictor for the development of PND (Belay et al., 2024). Additionally, the index created through PCA for social support is only a representation of household structure and aspects of marital structure. Variables assessing social networks were not provided in DHS datasets of selected countries.

Finally, and most importantly, this thesis only assessed the correlation between heat exposure and PND, as well as the lagged effects. While these insights are crucial, they do not allow for interpretation of causal pathways.

General Interpretation of Findings: Relation to Country Contexts

Findings carry important interpretations for selected countries. The protective effect observed in Bangladesh might be due to low temperature variability and a relatively high median temperature found in the data. Heat is not perceived as the greatest threat to health among environmental risk factors in

Bangladesh. The country experiences frequent and severe floodings that may take a greater toll on PND than heat exposure (Arafat & Rajkumar, 2024). Further, the protective effect may partly be explained by the timing of DHS fieldwork, which took place from July to December 2022, directly following severe flooding occurring between May and June (World Health Organization, 2022). With the fieldwork happening throughout a seasonal shift towards colder temperatures, this may have created a period of relative relief or recovery, that translated into the observed effect. Weems (2019) describes this phenomena as post-traumatic growth (PTG), which refers to a temporary uplift in wellbeing following crisis. Nevertheless, keeping in mind that the weighted prevalence estimate of PND for this country was high at 26%, another hypothesis might be valid; respondents from Bangladesh reported high PHQ-9 scores because of distress due to experienced floodings, even when maximum temperature was not higher than the median. After all, trauma due to extreme weather events, including floods, are associated with increased risk of depressive symptoms during the perinatal period (Barkin et al., 2024).

In Mozambique and Nepal, the most pronounced results have been observed, possibly due to highest variations in daily maximum temperatures. Mozambique was warmest on average, whereas Nepal showed the most extreme values (see Appendix 4, 5). In addition to high heat risk in Mozambique, the country is frequently hit by tropical cyclones and floodings. During the DHS fieldwork period, the longest tropical storm on record, “Freddy”, made landfall in Mozambique between February and March 2023. Reportedly, more than 1.3 million people were directly affected (WMO, 2024). This may, in part, contribute to the observed correlation. The results for Nepal were also showing a clear association, potentially due to the highest observed daily maximum temperature variability. Nepal’s diverse topography ranges from lowlands to alpine areas and contribute to this significant temperature variance. Rapid fluctuations were observed (see Appendix 5) which may potentially be a strong stressor for PND. In addition, the large temperature range allows for considerable contrast in exposures, which increases the sensitivity of the DLNM model to detect associations. Lastly, all forms of domestic violence were strong predictors of PND, as shown by chi-squared significance testing (See Table 2). This suggests a cumulative and mediated pathway from heat exposure through an increase in interpersonal violence to PND.

In Lesotho, a modest and insignificant association has been found. This may be due to a small sample size (only 69 women, contributing around 5% to the total study population), making it harder for statistical analyses to capture true effect estimates.

6 CONCLUSION

This thesis investigated the correlation between heat exposure and PND in 1,836 women from four LMICs to inform global and public health priorities.

The first research question explored the association between heat exposure and PND in Bangladesh, Lesotho, Mozambique, and Nepal. The observed correlations varied substantially across countries. Bangladesh was the only country where heat exposure (75th and 95th percentile) was associated with a protective effect on PND. Interestingly, lower maximum temperatures (5th and 25th percentile) demonstrated high and significantly elevated odds for PND. In Lesotho, temperature-related effects on PND were found to be weak and non-significant. Temperatures at the 5th and 25th percentiles are linked with slightly lower odds of PND compared to the reference temperature, while heat exposure is associated with modest but non-significant higher odds of PND. In Mozambique, a strong and positive association was found between heat exposure and PND, with temperatures above the 50th percentile showing increased odds of PND. Nepal also showed a strong association between exposure and outcome. Temperatures below the 50th percentile were protective, while temperatures above the median were associated with significantly higher odds of PND.

The second research question targeted the investigation of lagged effects between heat exposure and PND in selected countries. Results show country-specific and time-dependent effects of heat exposure on PND. In Bangladesh, temperatures above the 50th percentile showed highest odds for PND at lag 0 (OR = 1.19 (95%CI: 0.98–1.43)), and then remained consistently below 1.0, indicating that heat exposure has a protective effect across all lags. For Lesotho, odds for temperatures at the 75th and 95th percentiles were slightly elevated at lag 0 (OR = 1.05 (95%CI: 0.96–1.16)), suggesting an acute effect. However, estimates remained close to 1.0 and lacked statistical significance across all lags. This indicates no meaningful delayed effect of heat exposure on PND. In Mozambique, both acute and delayed effects of heat exposure were observed. The strongest heat-related odd was found at lag 1 (OR = 1.37, 95% CI: 1.07–1.75) for the 95th percentile, followed by a decline and then a secondary peak at lag 13. This suggests both an acute and delayed correlation in this setting. In Nepal, heat exposure was associated with acute effects, with ORs peaking at lag 0 (OR = 1.45, 95% CI: 1.11–1.89) for the 75th and 95th percentiles. After a decline until lag 5, the probability for PND rises again at longer lags, suggesting a delayed effect.

7 GLOBAL HEALTH PERSPECTIVES

This chapter provides a perspective on the relevance of this thesis to global health, first outlining further research needs and then broader implications.

Future Research Directions

To address the limitations and to fully understand the association between heat exposure and PND, further research is needed in this area. Findings should be validated using longitudinal rather than cross-sectional data to provide a rich and comprehensive understanding of the relationship. Assessing mental health at multiple time points, combined with continuous temperature exposure (both indoor and outdoor), offers great potential for exploring the dynamics of PND and individual-level factors. This approach also helps to eliminate ecological fallacy by ensuring that associations observed at the population level are confirmed by individual level data. In addition, research should explore the lived experiences of perinatal women with PND and the underlying pathophysiological mechanisms to fully understand the causal pathways. This triangulation of findings would contribute to a solid and multidimensional research base, guiding targeted intervention, public health preparedness, and climate-resilient maternal mental health policies.

Given the country-specific correlations and lagged effects observed, it is also critical to investigate the transferability of findings to other geographic and socioeconomic settings, particularly in LMICs that face disproportionate climate-related burdens (Buchwald & Boudova, 2024; KC & Kreyenbaum, 2025). Looking at the results from Bangladesh, if validated, there is an opportunity to gain insight into potential resilience that can inform mitigation and adaptation strategies across countries.

Additionally, future studies should also include other environmental stressors into statistical modelling – such as air pollution and humidity (Buchwald & Boudova, 2024) – and test different definitions of heat exposure. Significant inconsistencies between studies often come from the lack of a universally adaptable heatwave definition and the reliance on different exposure thresholds (Ulrich et al., 2025). Similarly, the lack of consensus on critical perinatal time windows of vulnerability further complicates comparisons between studies. Stratified analyses by pregnancy and postpartum status, depressive symptom severity as well as quantification of mediating and moderating factors are essential for the identification of high-risk groups and most appropriate timing of interventions. As Runkle and colleagues (2024) emphasize, evidence about the exact time window for delayed heat-related mental health effects remains inconclusive. While this thesis offered a wider lag assessment than the studies discussed, cross-validation in diverse local contexts is crucial for generalizability.

Broader Global Health Implications

Considering the emerging threat of climate change and heat exposure to PND, and possibly broader maternal mental health outcomes, immediate and targeted action is required. These efforts must adopt an intersectional approach to address the complex interplay of socio-demographic, socioeconomic, environmental and health system-related risk factors.

From a clinical perspective, Runkle et al. (2024) underscore the importance of expanding health care providers' awareness about the impacts of heat exposure beyond adverse birth outcomes to include the full spectrum of maternal mental health. Enabling them to act as multipliers for risk awareness increase offers a feasible, sustainable approach with little systemic barriers. This promotion of awareness could be integrated into routine care, such as antenatal care visits, to ensure access for perinatal women. This offers great potential, considering nurses are perceived as most trustworthy healthcare staff, highlighting their critical role in risk communication (Barkin et al., 2024).

Despite growing relevance of health consequences of climate change, funding for climate-related mental health research remains extremely limited, particularly in the context of maternal health. To streamline financial resources into this area, elevating risk awareness about the long-lasting, broad health impacts and associated costs as well as intersections with other public health priorities (e.g. non-communicable diseases, infectious diseases) may help to attract the attention of policymakers and funders to this critical issue.

Heat exposure and PND closely align with the UN's Sustainable Development Agenda, Goal 3 about "Good Health and Well-Being for All". This goal must move beyond the mere focus on maternal mortality which has long been the primary target and indicator of global public health programs (Handa et al., 2024). A shift is needed in global health priorities, moving to a more holistic perspective that includes mental well-being as a core indicator of quality of life, not just for women, but also for their offspring and society. After all, addressing this gap is not just a matter of advancing science, but also of women's rights and human rights, social and climate justice, as well as health equity (Buchwald & Boudova, 2024).

DECLARATION OF AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, I used ChatGPT to assist with coding in Stata, R, and Python, as well as linguistic improvements. Additionally, DeepL was utilised for grammatical control and assistance in rephrasing. After using these tools, I reviewed and edited the content as needed and take full responsibility for the content of the work.

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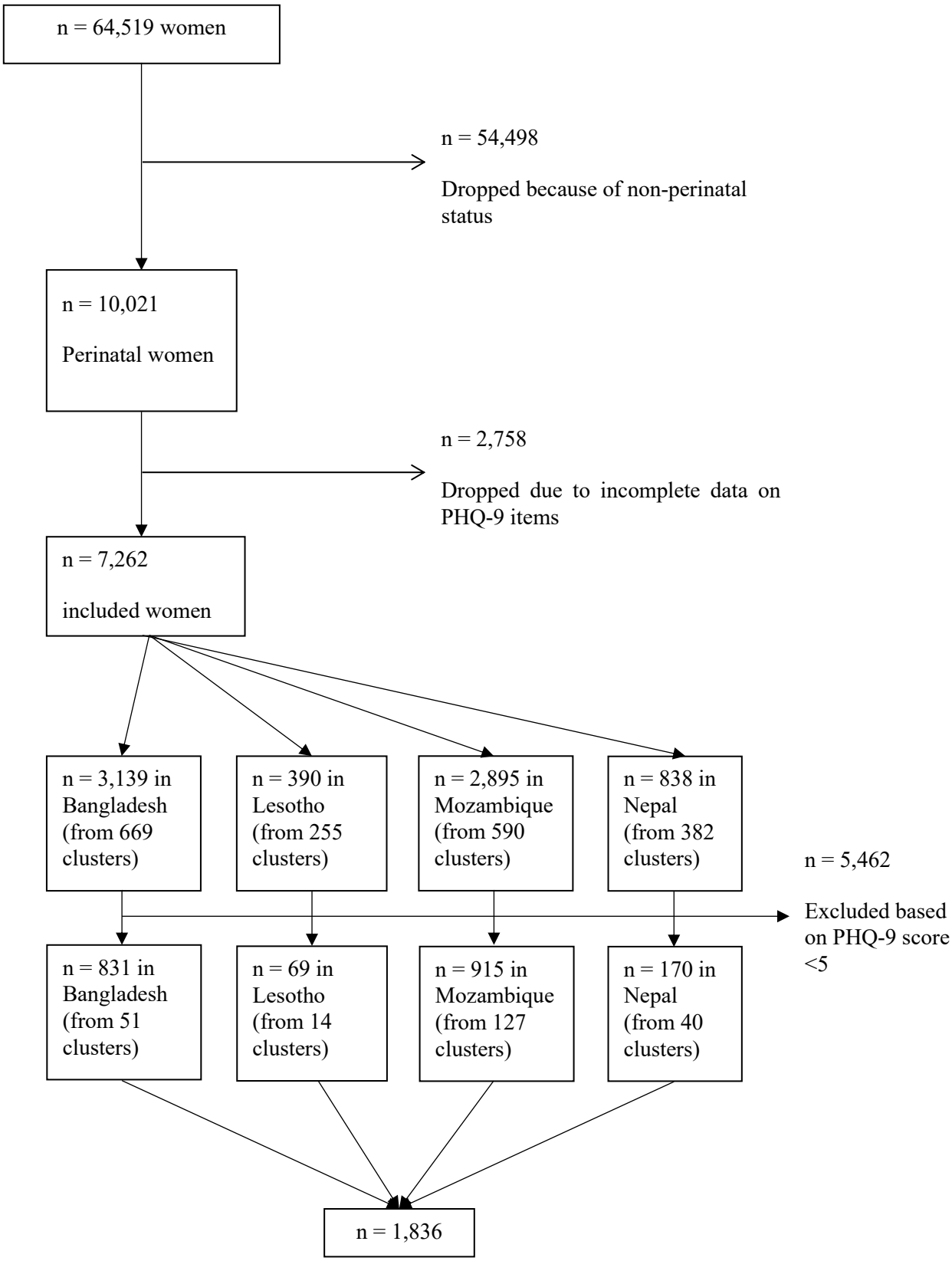
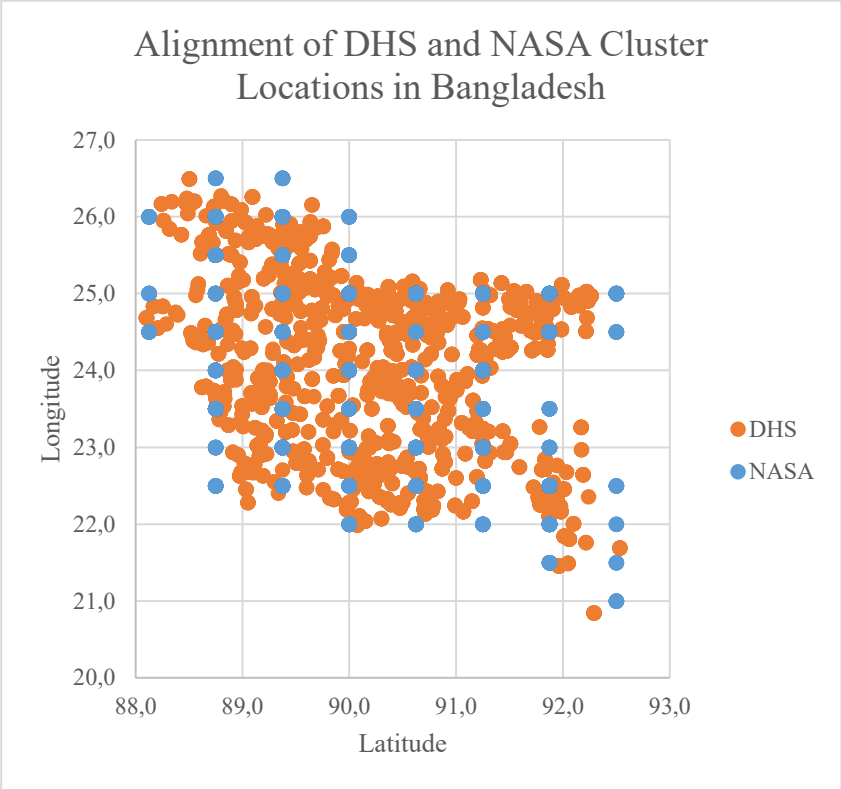
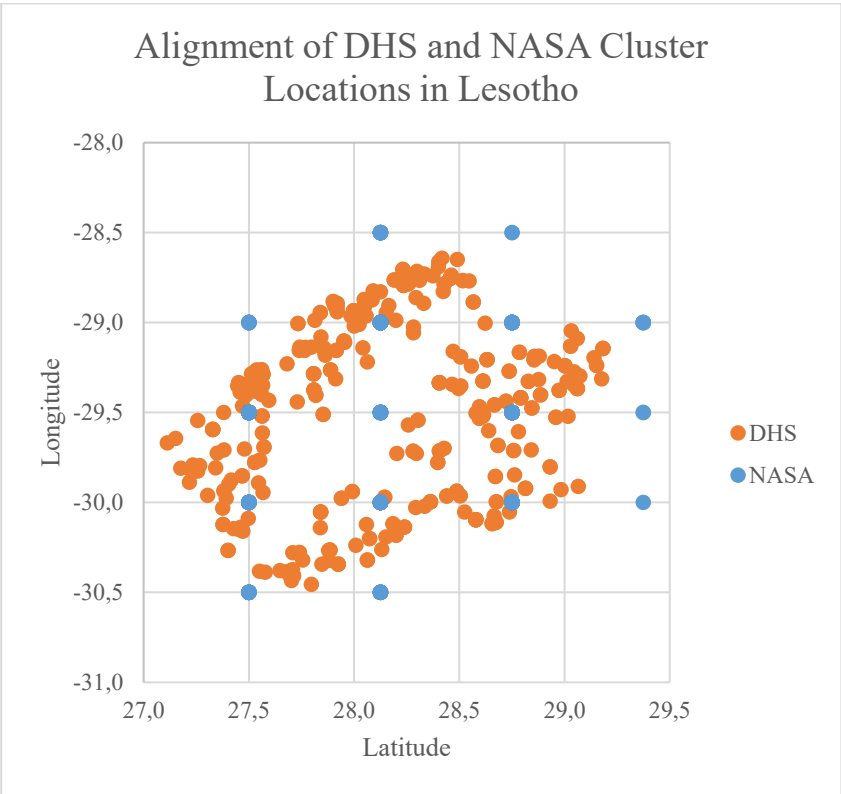
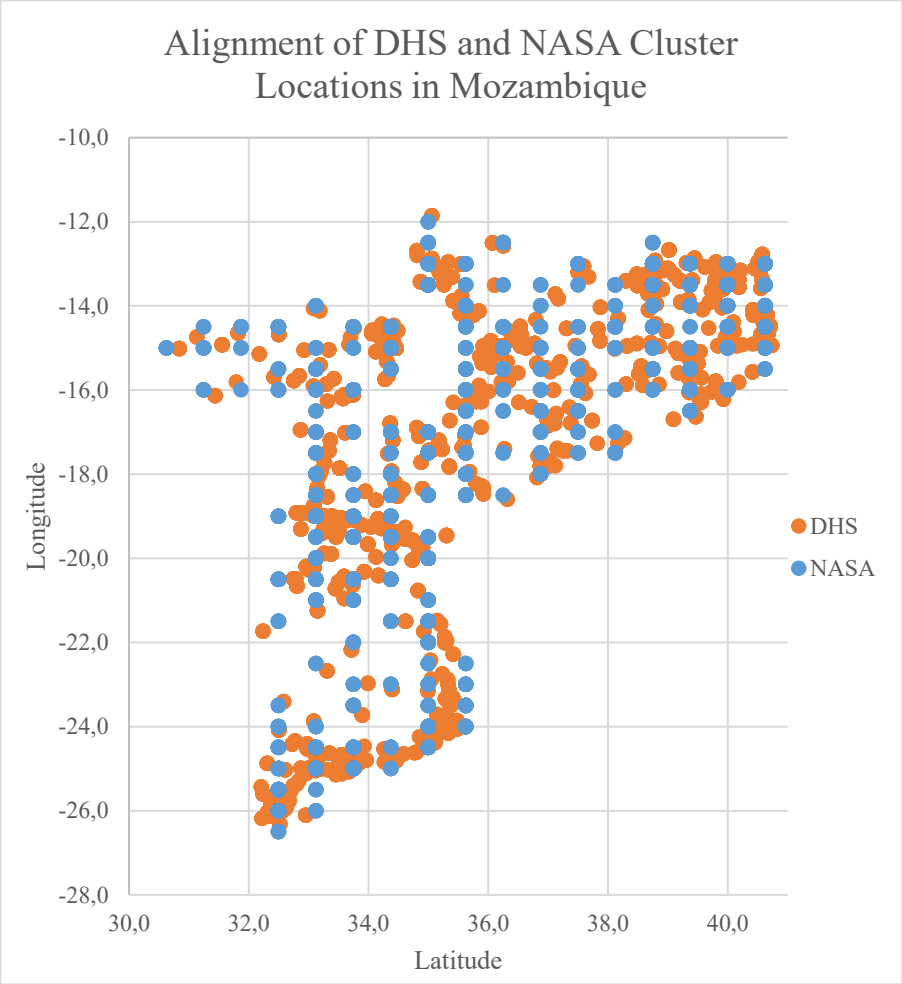
Appendix 1. *Flow Chart of Participants*

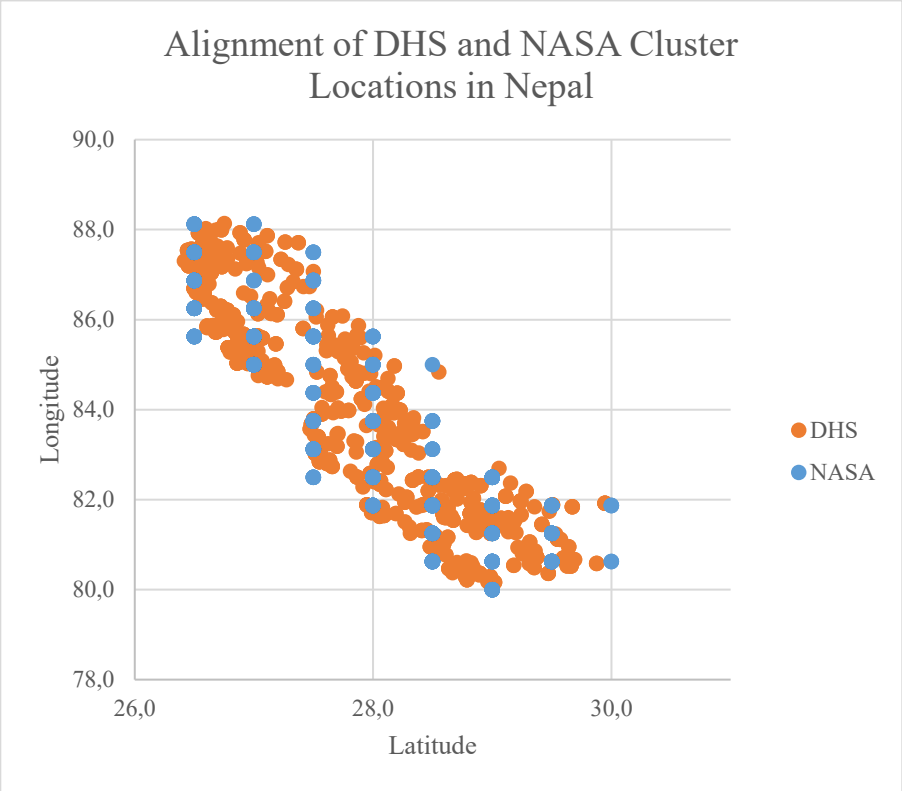


***Appendix 2**. Spatial Alignment of NASA Grid with DHS Cluster Coordinates*






Alignment of DHS and NASA Cluster Locations in Mozambique
-10,0
-12,0
-14,0
-16,0
-18,0
-20,0
-22,0
-24,0
-26,0
-28,0
Longitude
30,0
32,0
34,0
36,0
38,0
40,0
Latitude
DHS
NASA

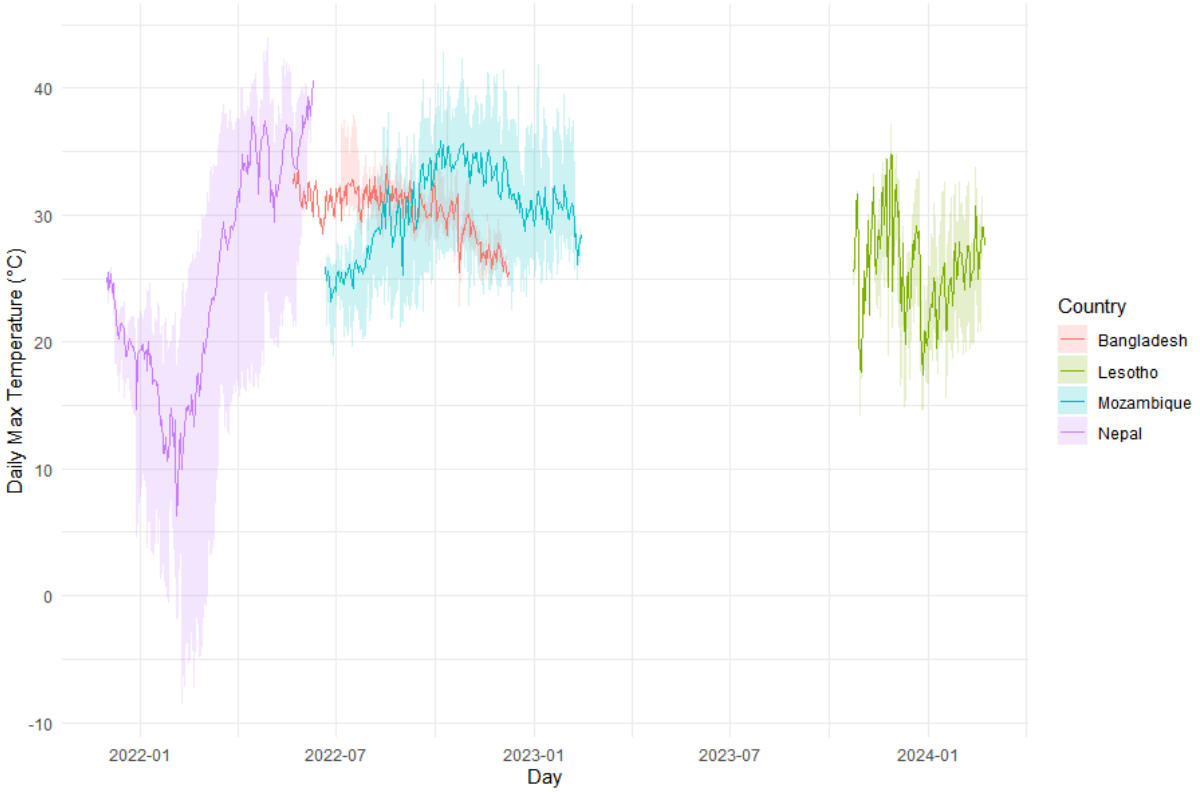

Alignment of DHS and NASA Cluster Locations in Nepal
90,0
88,0
86,0
84,0
82,0
80,0
78,0
Longitude
26,0
28,0
30,0
Latitude
DHS
NASA

Appendix 3. *Number of DHS and NASA cluster Locations for Eligible Women and PND Positive Women*

	Bangladesh	Lesotho	Mozambique	Nepal
Eligible women[a] – DHS cluster	669	255	590	382
Eligible women[a] – NASA cluster	57	16	166	44
PND positive women[b] – DHS cluster	444	56	348	140
PND positive women[b] – NASA cluster	51	14	127	40

a: Sample size of 7,262
b: Sample size of 1,836

Appendix 4. *Graph of Average Daily Maximum Temperature (including Range) across all Clusters for PND Positive Women during DHS-8 Fieldwork in Bangladesh, Lesotho, Mozambique, Nepal*



Appendix 5. *Summary Statistics for Daily Maximum Temperature Across Countries in °C*

Country	Mean	SD[a]	Minimum	Maximum	Median	IQR[b]
Bangladesh	30.3	1.9	22.7	38.5	30.4	2.7
Lesotho	25.6	4.7	14.2	37.2	25.7	7.4
Mozambique	31.1	4.0	18.8	42.9	30.8	6.1
Nepal	21.6	9.8	-8.6	44.0	20.4	13.4

a: Standard Deviation

b: Interquartile Range

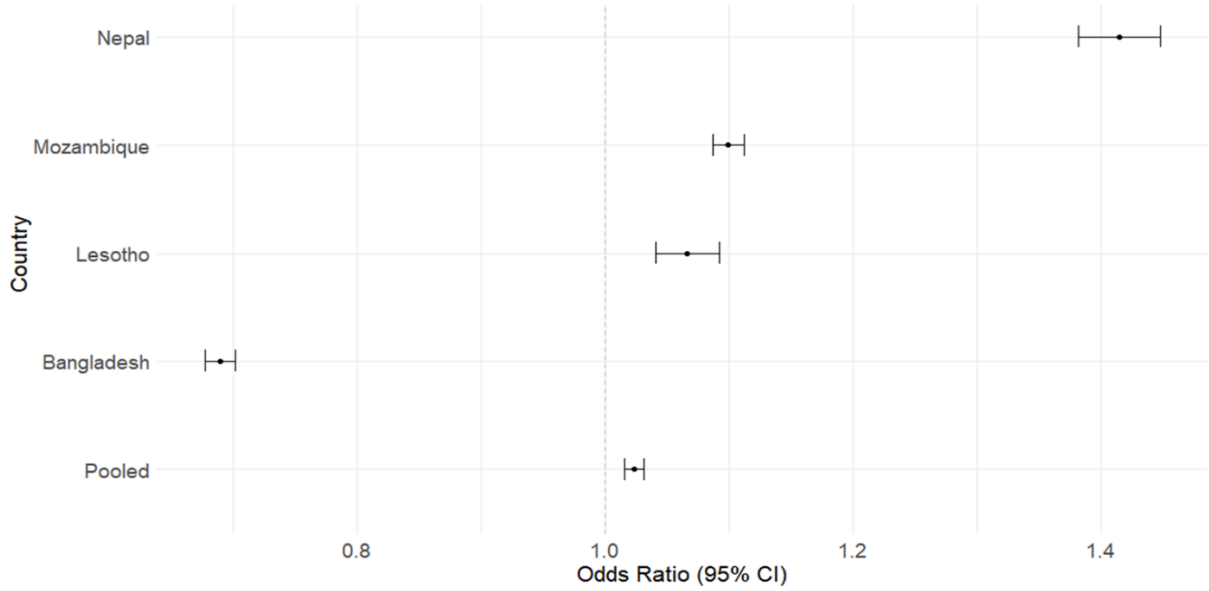
Appendix 6. *Odds Ratio and 95% CI from Conditional Logistic Regression, Long Format*

Country	Pseudo R^2	Std. Err.[b]	Odds Ratio	95% CI	p-value[a]
Bangladesh	0.0396	0.0060	0.69	[0.677, 0.701]	<0.001
Lesotho	0.0072	0.0131	1.07	[1.041, 1.092]	<0.001
Mozambique	0.0059	0.0065	1.10	[1.087, 1.112]	<0.001
Nepal	0.0999	0.0167	1.41	[1.382, 1.448]	<0.001
All countries[a]	0.0003	0.0040	1.02	[1.015, 1.031]	<0.001

a: n = 1,836. b: Significance level = $p < 0.005$

b: Standard Error

Appendix 7. *Conditional Logistic Regression performed in Long Format, showing Results for Correlation between Daily Maximum Temperature and PND (ORs and 95% CIs)*

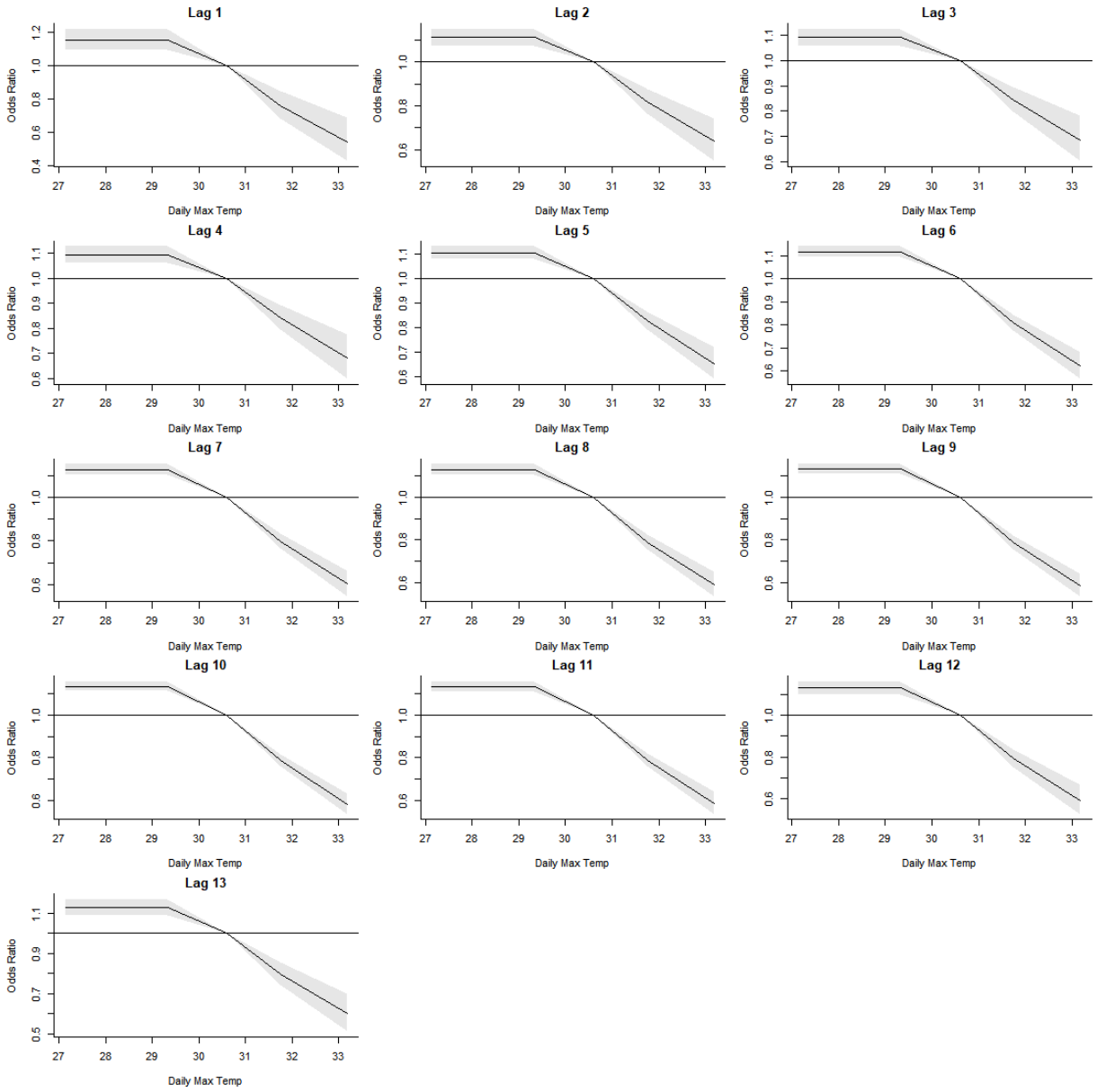


Appendix 8. *Lag-specific Odds Ratio and 95%CI for PND from Conditional Logistic Regression, Wide Format*

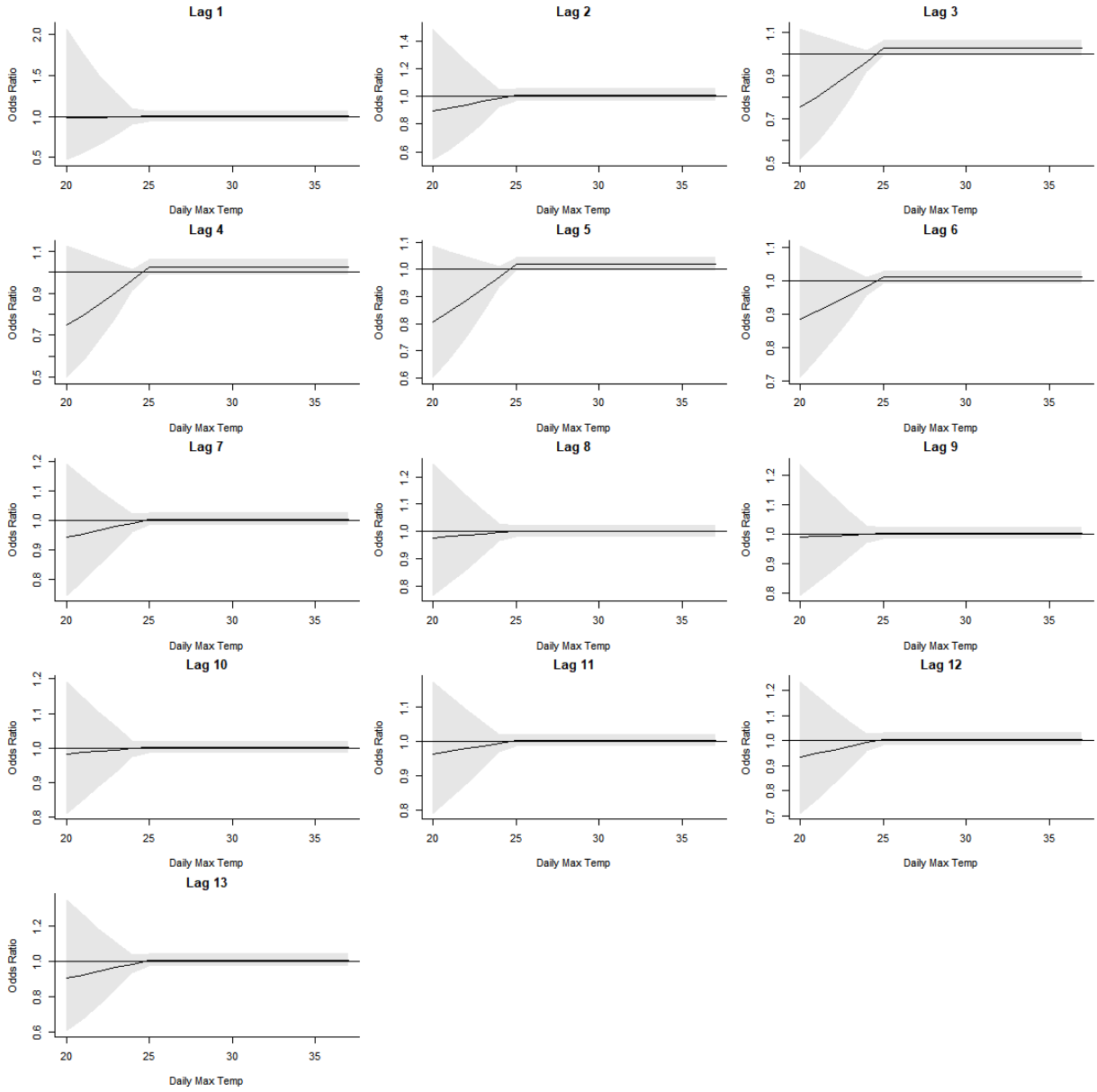
Lag	Bangladesh	Lesotho	Mozambique	Nepal	Pooled
Lag 1	0.75 (0.71–0.8)	1.07 (0.98–1.16)	1.14 (1.09–1.19)	1.29 (1.19–1.39)	1.04 (1.01–1.07)
Lag 2	0.76 (0.71–0.8)	1.07 (0.99–1.17)	1.09 (1.04–1.14)	1.36 (1.25–1.47)	1.03 (1–1.06)
Lag 3	0.69 (0.65–0.74)	1.05 (0.97–1.15)	1.08 (1.03–1.13)	1.37 (1.27–1.49)	1.01 (0.98–1.04)
Lag 4	0.67 (0.63–0.72)	1.04 (0.96–1.13)	1.1 (1.05–1.15)	1.40 (1.29–1.52)	1.01 (0.98–1.03)

Lag 5	0.66 (0.62–0.7)	1.03 (0.95–1.12)	1.11 (1.07–1.16)	1.33 (1.23–1.43)	1.00 (0.97–1.03)
Lag 6	0.61 (0.57–0.65)	1.04 (0.96–1.12)	1.1 (1.06–1.15)	1.35 (1.25–1.47)	0.99 (0.96–1.02)
Lag 7	0.67 (0.63–0.71)	1.08 (1–1.18)	1.12 (1.07–1.17)	1.34 (1.23–1.45)	1.01 (0.99–1.04)
Lag 8	0.72 (0.68–0.76)	1.11 (1.02–1.2)	1.11 (1.06–1.16)	1.34 (1.23–1.46)	1.02 (0.99–1.05)
Lag 9	0.74 (0.7–0.79)	1.06 (0.98–1.16)	1.1 (1.05–1.15)	1.35 (1.24–1.47)	1.03 (1–1.06)
Lag 10	0.73 (0.69–0.78)	1.04 (0.95–1.13)	1.1 (1.05–1.15)	1.36 (1.25–1.48)	1.02 (0.99–1.05)
Lag 11	0.75 (0.71–0.8)	1.04 (0.94–1.14)	1.08 (1.04–1.13)	1.56 (1.41–1.73)	1.03 (1–1.06)
Lag 12	0.72 (0.68–0.77)	1.04 (0.95–1.14)	1.11 (1.06–1.16)	1.45 (1.32–1.59)	1.04 (1.01–1.07)
Lag 13	0.74 (0.7–0.79)	1.06 (0.97–1.16)	1.11 (1.06–1.16)	1.46 (1.33–1.6)	1.05 (1.02–1.08)
Lag 14	0.72 (0.68–0.77)	1.11 (1.01–1.22)	1.07 (1.03–1.11)	1.47 (1.34–1.61)	1.03 (1–1.05)

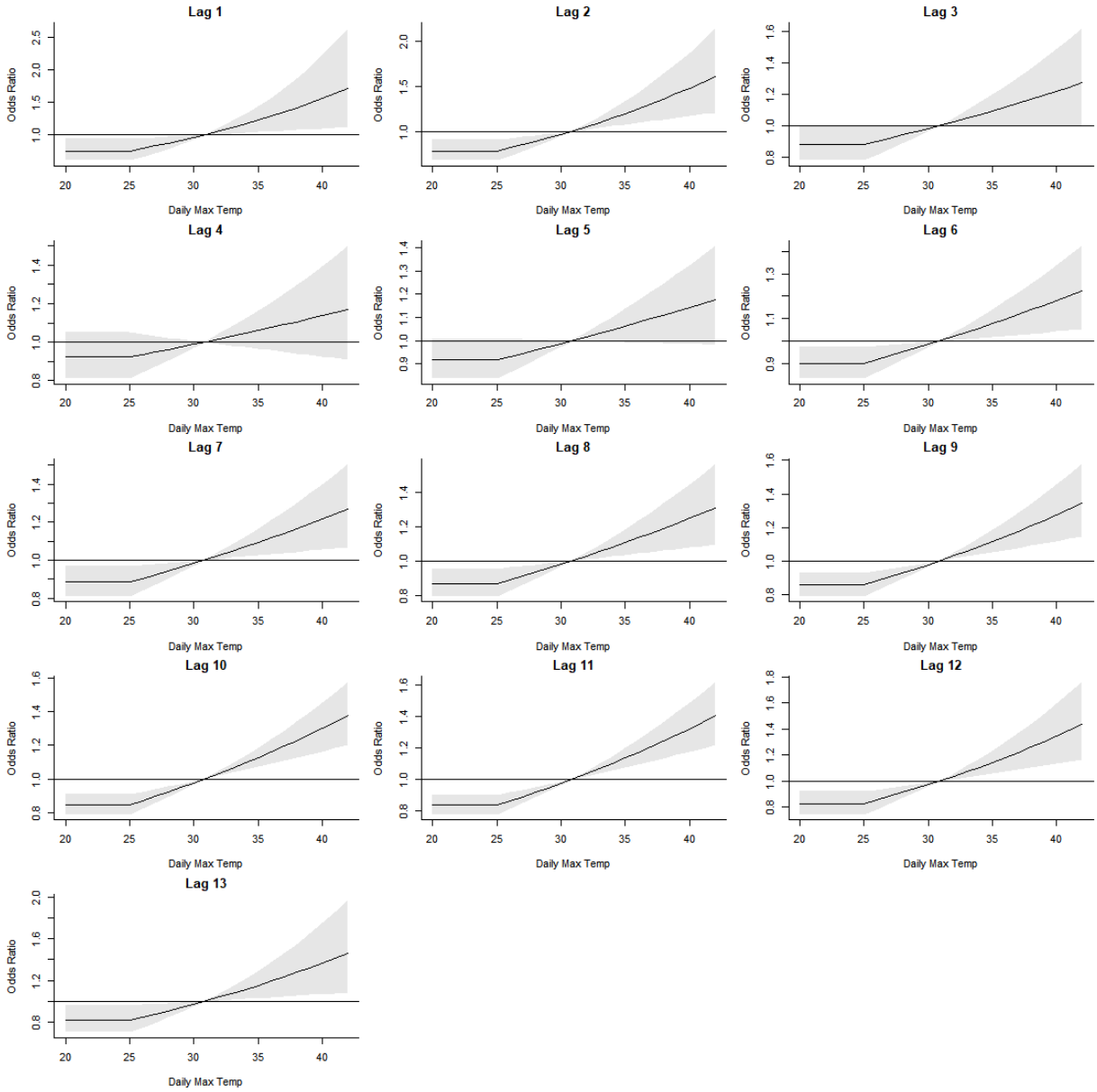
Appendix 9. *Slice Model Showing Lag-Specific ORs (95%CI) with Exposure, Bangladesh*



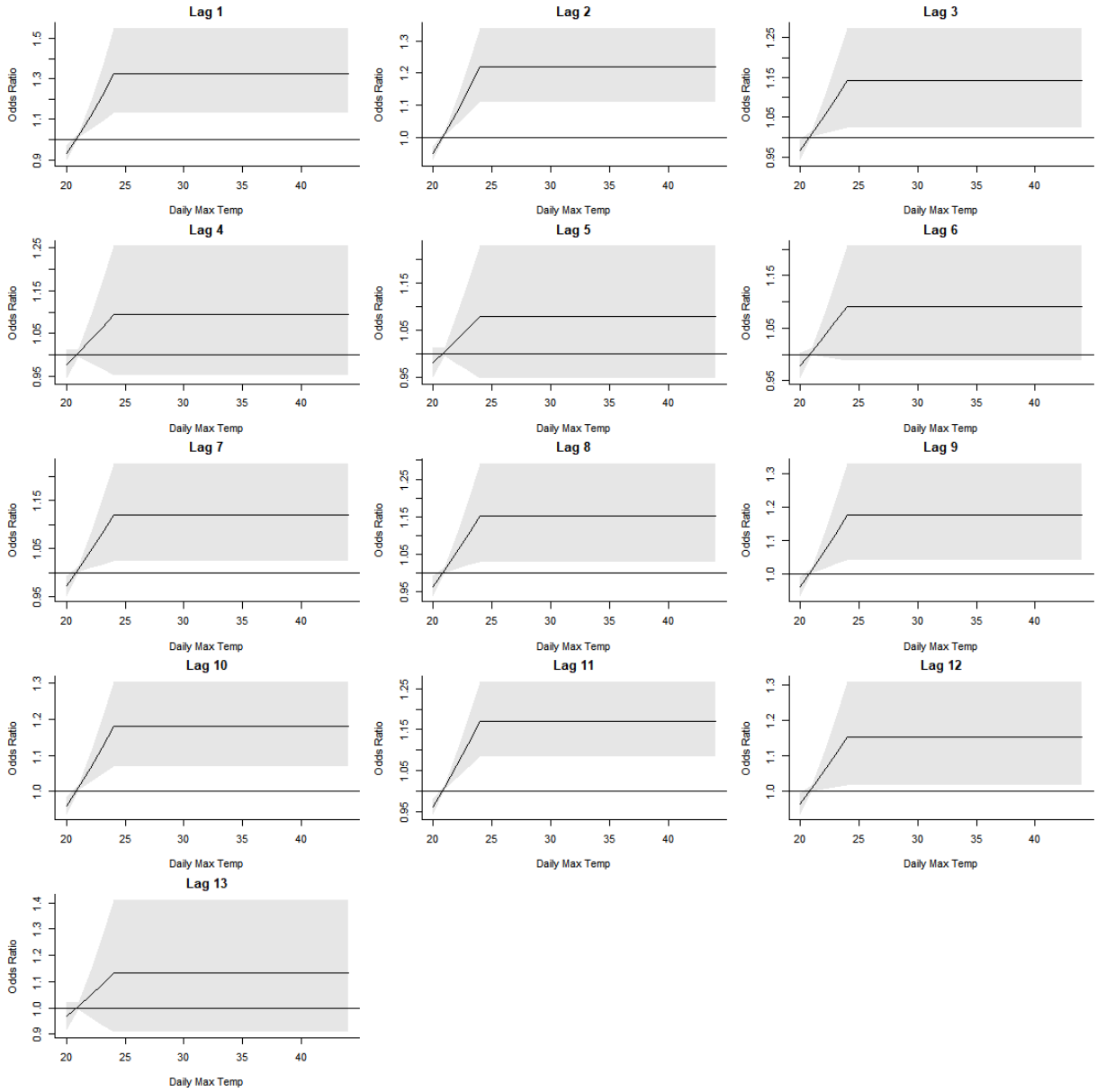
Appendix 10. *Slice Model Showing Lag-Specific ORs (95%CI) with Exposure, Lesotho*



Appendix 11. *Slice Model Showing Lag-Specific ORs (95%CI) with Exposure, Mozambique*

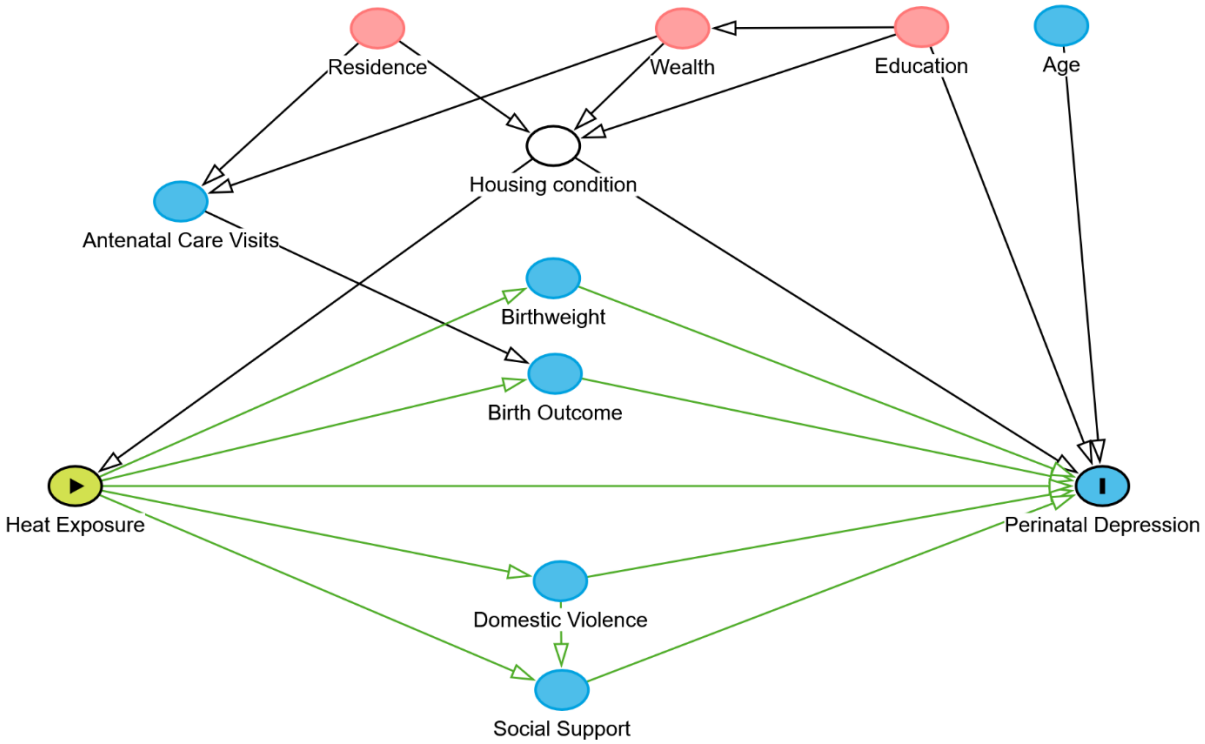


Appendix 12. *Slice Model Showing Lag-Specific ORs (95%CI) with Exposure, Nepal*



Appendix 13. *Systematic Literature Search Conducted in PubMed, 4th of February 2025*

#	Search Details	Results	Time
7	#4, Filters: Clinical Study, Clinical Trial, Comparative Study, Editorial, Meta-Analysis, Observational Study, Review, Systematic Review, English, Humans.	144	06:12:33
6	#4, Filters: English, Humans, from 2015 - 2025	416	06:08:42

5	#4, Filters: English, Humans	775	06:07:21
4	#1 AND #2 AND #3	1,485	06:05:48
3	"Depression, Postpartum"[MeSH] OR "Psychological Stress"[MeSH] OR "Anxiety Disorders"[MeSH] OR "Depressive Disorders"[MeSH] OR "Mental Disorders"[MeSH] OR "Perinatal depression"[All Fields] OR "Postpartum Depression"[All Fields] OR "Maternal Mental Health"[All Fields] OR "Perinatal Mental Health"[All Fields] OR "Maternal Stress"[All Fields] OR "Pregnancy Anxiety"[All Fields] OR "pregnancy depression"[All Fields] OR "perinatal anxiety"[All Fields] OR "postnatal depression"[All Fields] OR "antenatal depression"[All Fields] OR "maternal anxiety"[All Fields] OR "prenatal anxiety"[All Fields]	1,720,319	06:04:22
2	"Hot Temperature"[MeSH] OR "Extreme Heat"[MeSH] OR "Climate Change"[MeSH] OR "Global Warming"[MeSH] OR "heatwave"[All Fields] OR "heat waves"[All Fields] OR "heat"[All Fields] OR "extreme heat"[All Fields] OR "high temperature"[All Fields] OR "hot temperature"[All Fields] OR "ambient temperature"[All Fields] OR "heat exposure"[All Fields] OR "temperature stress"[All Fields] OR "thermal stress"[All Fields] OR (non-optimal temperature)	1,015,591	05:56:11
1	"Pregnancy"[MeSH] OR "Postpartum Period"[MeSH] OR "Pregnant Women"[MeSH] OR "Peripartum Period"[MeSH] OR "Maternal Health"[MeSH] OR "pregnancy"[All Fields] OR "prenatal"[All Fields] OR "antenatal"[All Fields] OR "perinatal"[All Fields] OR "postpartum"[All Fields] OR "maternal health"[All Fields] OR "pregnant women"[All Fields] OR "new mothers"[All Fields] OR "peripartum period"[All Fields] OR "postnatal period"[All Fields] OR "gestation"[All Fields]	1,460,376	05:45:55

Appendix 14. *Directed-Acyclic-Graph (DAG) of Assumed Pathway between Heat Exposure and Perinatal Depression*